

**REGTECH: AUTOMATING FINANCIAL COMPLIANCE USING
AI AND BLOCKCHAIN FOR REAL-TIME RISK MONITORING**

Candidate's Declaration

I assert that this dissertation and the work contained within are my accomplishment. Where I have drawn on the published literature of others, this is always acknowledged; Where I have borrowed directly from the work of others, the source is indicated. With the only exception of such quotations, this dissertation represents my work; I have thanked everyone who helped me; I have familiarised myself with the sanctions of Academic Misconduct.

Signed:_____

Date:_____

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Abstract

The dynamic financial environment presents many regulatory compliance challenges as online financial transactions are becoming complex and regulatory input is also changing very rapidly. This dissertation examines the conceptualisation of an intelligent Regulatory Technology (RegTech) prototype that combines **Artificial Intelligence (AI)** and blockchain to automate regulatory processes in the banking sector and make real-time risk monitoring possible. Logistic Regression, Random Forest, and Isolation Forest models were trained on secondary data of a financial transaction data to provide the protective measures that risk or do not risk: a transaction. The models were tested in terms of typical classification measurements, including **accuracy, precision, recall, F1-score and ROC-AUC**, with Random Forest performing better in predicting anomalous patterns.

In order to maximise auditability and assure data integrity, a permissioned Ethereum-based blockchain was adopted using a custom Solidity-based smart contract. The blockchain element captures the flagged transactions in such a way that they are immutable, and the recorded line of information includes transaction IDs, risk scores, and timestamps in the Web3-powered interface. A **command-line interface (CLI)** was implemented that makes the logging process seamless by associating machine learning predictions with the blockchain storage. This architecture guarantees that every high-risk transaction that is detected by the AI system is openly and permanently documented, thus allowing secure regulatory scrutiny.

The methodology used in the project is a hybrid that incorporates **CRISP-DM** to be used in data mining, **SDLC** to be used in system development, and Agile design, which considers iterative design. Data privacy and model fairness, as well as other Ethical considerations, were built into the model. The proposed solution exemplifies the future-proof and explainable aspect of financial compliance and provides an innovative efficiency, precision, and non-disguised nature as compared with the conventional way of compliance. Our study will add to the already burgeoning literature on RegTech in the following ways: (1) it will introduce a working prototype that realizes the possibilities of better integrating predictive analytics and regulatory accountability through secure, decentralized systems; (2) it will provide an implicit comparative analysis of the viability of the said possibilities, and (3) it will extend the literature by proposing a specific navigation matrix frame based on their functional implementation.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
RegTech	Regulatory Technology
FinTech	Financial Technology
KYC	Know Your Customer
AML	Anti-Money Laundering
GDPR	General Data Protection Regulation
SDLC	Software Development Life Cycle
CRISP-DM	Cross Industry Standard Process for Data Mining
CLI	Command-Line Interface
API	Application Programming Interface
ABI	Application Binary Interface
JSON	JavaScript Object Notation
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
XAI	Explainable Artificial Intelligence
ETH	Ethereum
CSV	Comma-Separated Values
UI/UX	User Interface / User Experience
ID	Identifier
Txn	Transaction
GANACHE	Ethereum Blockchain Local Development Tool
EVM	Ethereum Virtual Machine
F1-Score	Harmonic Mean of Precision and Recall

Chapter 1: Introduction

1.1 Overview

The digital financial transactions have been growing exponentially, putting pressure on digital financial systems to be compliant with automation. **Regulatory Technology (RegTech)** has become a strategic option, with the aim of easing the burden of regulatory compliance by means of integrating advanced technologies. Traditional RegTech solutions can be non-adaptable in real-time and lack auditability to deal with the dynamic nature of compliance. This research paper is a proposal for the creation of a RegTech prototype platform that will utilise supervising machine learning algorithms to detect anomalies and utilise blockchain architecture to have an unalterable audit. Incorporating AI with blockchain into one integrated system, the study would allow ensuring real-time risk detection and improving the transparency of compliance procedures. The research uses the secondary data collected and publicly available in anonymised form of financial transactions data and applies Python-based technologies in its implementation. Its higher goal is to determine whether such an integrated system can be developed and how efficient it can be in the development of compliance assurance within the financial sphere.

1.2 Rationale

Financial transactions are more numerous, complex and unpredictable than ever before, and the regulatory environment continues to change and become more complex. The need to move away from the traditional, manual means of handling compliance and adopt a new approach to process financial transactions utilising intelligent and automated remedies is a paradigm shift that is visibly needed. Financial processes and institutions are under pressure to execute to high standards such as **Anti-Money Laundering (AML)**, **Know Your Customer (KYC)** and data privacy regulations, not to mention the risk of fraud and systemic collapse risk. Currently deployed RegTech-based systems, although digitised, tend to reuse older processes and are not as real-time as risk analysis, audit requirements, or scalable.

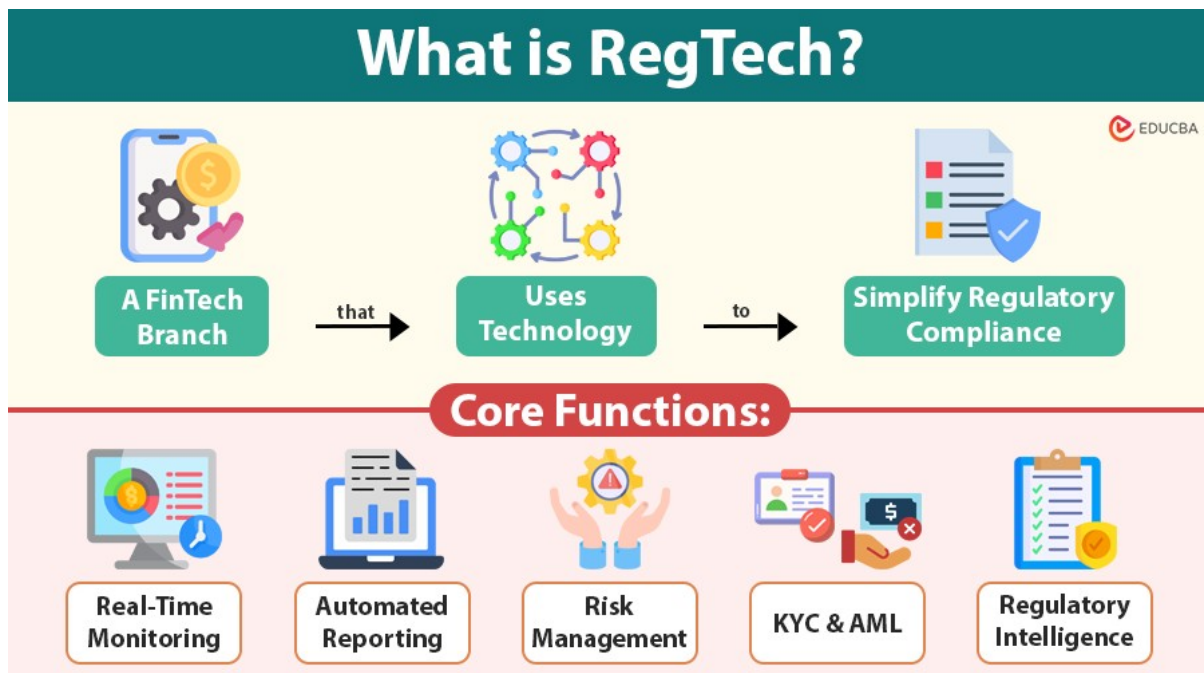


Figure 1: RegTech Information

(Source: educba.com)

The combination of **Artificial Intelligence (AI)** and Blockchain technologies has the potential transformative potential in improving regulatory compliance. AI facilitates the dynamic finding of suspicious transactions through anomaly detection, and Blockchain will also allow data immutability, transparency, and decentralised audit trails. The combination of such technologies can help in the creation of a resilient compliance framework that is reactive as well as predictive in nature.

The study pursues a semi-qualitative approach to the analysis and development of a Python-based RegTech prototype with the use of anonymised financial data. This will turn into a prospect of combined experimentation and testing machine learning models and blockchain integration without jeopardising sensitive personal information (Kothandapani., 2024). The justification, therefore, becomes creating a scalable, safe and smart compliance automation tool, serving as a solution to the problems in terms of inefficiency in operations and regulatory overburden in the banking sector.

1.3 Aim and Objectives

The core objective of this research is the design, development and evaluation of a RegTech prototype that combines Artificial Intelligence (AI) and Blockchain to automate financial

compliance operations and to monitor risks in real-time within the banking industry. This prototype will use supervised machine learning-based anomaly detection methods and apply a permissioned blockchain-based infrastructure to permissions to log regulatory activities and hence enhance visibility, efficiency, and clear accounting of compliance procedures.

To realise this goal, it has set the following objectives:

- To question the shortcomings of traditional financial compliance approaches, the RegTech solutions available, and the inefficiencies of dealing with complex regulatory problems.
- To research and develop AI algorithms with Python (e.g., Random Forest, Logistic Regression, Isolation Forest) to classify anomalous or suspicious transactions in financial data and detect these instances.
- To design a permissioned blockchain system using Ethereum (either on Ganache or via Remix) as the vehicle to record compliance-related events and maintain traceability and the integrity of audit trails.
- To connect the AI detection element with the blockchain ledger to develop an automated, end-to-end compliance verification system.
- To compare the performance of the prototype system to conventional compliance techniques for accuracy, speed, transparency and ease of regulatory audit.

1.4 Research Questions

The question that influences this research is to identify the possibility of combining Artificial Intelligence (AI) and blockchain technology in order to improve regulatory compliance in the banking industry. With the drawbacks of the traditional RegTech models and the increasingly sophisticated regulatory environment of financial institutions, the paper aims to answer the following research questions:

1. How can intelligent algorithms, especially the models of the supervised machine learning type, be employed to identify the compliance risks and anomalous financial transactions in real time?
2. What is the viability of using blockchain technology to make sure that compliance event logging is immutable, traceable, and transparent?

3. What are the drawbacks of the proposed AI-blockchain hybrid RegTech framework relative to a traditional compliance monitoring system in terms of efficiency, auditability and scalability?
4. What are some of the principles of design and practical implementation issues in the design of a secure and GDPR-compliant RegTech prototype provided with secondary financial transaction datasets?

Such questions will support the methodology and evaluation approaches followed in the study.

1.5 Structure of the Dissertation

The dissertation is organised systematically so as to provide a logical flow of theoretical underpinnings and their application into practice, as well as appraisal. Chapter 1 gives an overview of the study, which states the rationale, aim and objectives of the study. Chapter 2 is an evaluative literature review of the published literature pertaining to RegTech, financial compliance, and the synthesis of artificial intelligence and blockchain technologies. The methodology that has been adopted includes the use of the CRISP-DM framework of data-driven modelling, the SDLC approach to creating a system and the Agile method of iterative prototyping. Chapter 4 provides the architectural design and implementation of the proposed RegTech prototype. Chapter 5 provides the results of the AI models and blockchain system, such as an evaluation of all the performances. Chapter 6 is devoted to project management practices, such as the setting of the schedule, resource management, and risk elimination course of action. Chapter 7 completes the dissertation by concluding with a summary of the key findings, contributions, limitations, and recommendations with respect to any future research work and development.

1.6 Summary

This chapter has framed the research topic, stated the rationale and given a description of the main aim and objectives driving the research. The growing complexity of financial regulation, as well as the inefficiency of conventional compliance mechanisms, proves the necessity of a computerized and intelligent mechanism. Using artificial intelligence to detect risk in real-time, as well as by using blockchain to have secure audit trails, a prototype RegTech framework should be developed. The chapter also outlined the research questions and structural roadmap,

and this formed the basis of the subsequent literature research, methodological framework, system design, empirical performance, and final analysis.

Chapter 2: Literature Review

2.1 Introduction

The transformation of the financial environment has created the need for more solid, versatile and transparent regulatory compliance mechanisms. **Regulatory Technology (RegTech)** is becoming an essential subsection of FinTech, which serves to digitise and robotise the compliance infrastructures to counter the rising regulatory complexities and operational risks. Yet, numerous current RegTech tools are restrictive in their use, often as nothing more than manual compliance procedures automated by simple digital interfaces and not allowing to enable real-time monitoring and intelligent decisions. As financial transactions continue to grow exponentially and obligations of compliance are highly dynamic, conventional systems are becoming manifestly inefficient. This literature review critically assesses the process of integrating **Artificial Intelligence (AI)** and Blockchain technologies in the RegTech sector for automation financial compliance and better managing risks. The review critically appraises peer-reviewed academic and industry literature to discover new technological achievements and barriers to implementation, as well as research gaps. It would be aimed at offering a conceptual underpinning of the development of an intelligent, secure, and scalable RegTech framework with the aim of targeting the banking domain.

2.2 Overview of Financial Compliance in Banking

According to Olaiya *et al.*,2024, financial compliance in banking is the orderly conformity with regulatory requirements, work standards and statutory guidelines laid down by national and international financial regulators. The compliance measures will aim to maintain the integrity, transparency, and legal conduct of the financial institutions that are acting in the interest of their customers and ensuring systemic stability. In response to greater globalisation, digitisation of banking services and the sheer growth of financial crimes like money laundering, terrorist funding and tax evasion, the regulatory landscape has become more dynamic over the past 20 years.

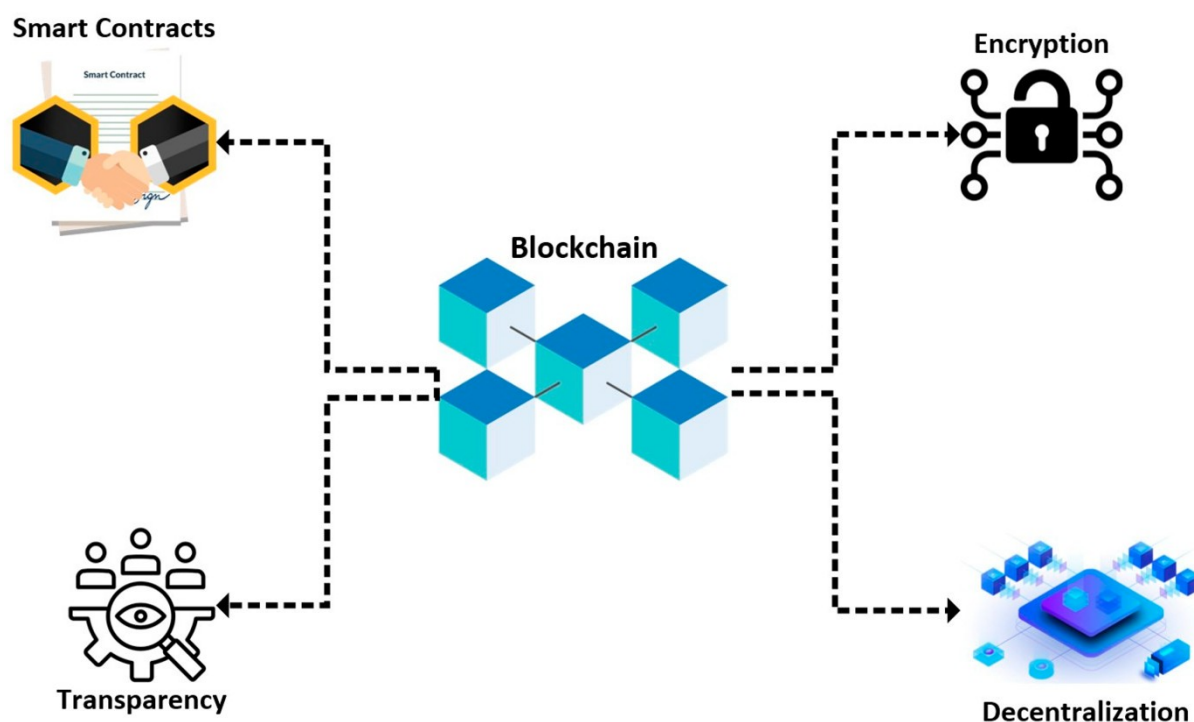


Figure 2: Enhancing Decentralised Decision Making with Big Data and Blockchain Technology

(Source: mdpi.co)

Financial institutions in modern times are required to meet a set of regulatory requirements comprising **Know Your Customer (KYC)**, **Anti-Money Laundering (AML)**, and **General Data Protection Regulation (GDPR)**. The directives mandate the banks to implement effective systems of internal controls, conduct a continuous risk assessment, conduct due diligence of customers and implement audio and virtual paths that are secure and verifiable. Failure to comply may lead to significant monetary fines, loss of reputation and limitation of operations.

Financial compliance is comprised of core capabilities to include real-time transaction monitoring, suspicious activity reporting, regulatory reporting and data privacy enforcement. Such processes were traditionally handled using manual procedures, spreadsheets, and legacy software tools, which are generally seen as inefficient, riddled with errors, and ill-suited to deal with the complexity and huge volume of contemporary financial data.

As cross-border banking and the use of digital financial services have increased, compliance operations have grown more functionally diverse and complicated (Kodete., 2021). Financial institutions are being compelled to become more adaptive and technology-driven in ways of the continuous compliance and management of risks. As such, the combination of technologies,

including **Artificial Intelligence (AI)**, **Natural Language Processing (NLP)**, and **Blockchain**, is becoming a key strategic initiative to automate compliance processes, increase accuracy, and ensure auditability. This move signifies a paradigm change in the way compliance is understood- not as a regulatory demand but as an important dimension within the contexts of digital banking environments.

2.3 Challenges in Traditional Compliance Methods

The conventional approaches to compliance, in the banking and financial services sector, are greatly operationally and technically constrained from working effectively to conform to the emerging compliance requirements. Such approaches are much more manual-based data entry, rules-driven audits, and locked-down reporting systems that are ill-suited to the volume, velocity and variety of transactions in modern finance. With the continued evolution of regulatory environments that have increasingly become more stringent, legacy systems have a limited ability to adapt to these changes in management in real-time and therefore lead to a lag in anomaly detection and exposure to compliance risks.

The major drawback of the traditional solutions is the lack of automation and the lack of real-time processing. The majority of compliance checks are performed after the fact, which limits the potential of such checks in terms of risk mitigation. Manual tasks will also make errors and inconsistencies more likely to occur, especially in large-scale operations having high levels of transaction frequency. Moreover, these systems tend to be used in silos, without connection to other external data sources, which limits their capability of delivering in-depth insights of compliance.

In addition, traditional compliance frameworks that are traditional lack scalability and flexibility. They cannot operate dynamically to reflect changes in regulatory requirements or adjust themselves to changing jurisdictional requirements without heavy manual adjustment (Rane *et al.*, 2023). This inflexibility adds to the overhead and allocation costs of resources, particularly when there is a regulatory audit or due to a reporting cycle.

Data transparency and traceability are another issue of concern. A paper-based or non-standardised electronic files lack immutability and the ability to verify a historical audit trail, and so regulators and internal auditors cannot easily confirm the compliance activities of the

past. Moreover, conventional systems are poorly responsive to complex financial crimes, including money laundering, because patterns cannot be recognised or learning is not adaptive.

These weaknesses taken together call for a paradigm shift to more intelligent, technology-driven models of compliance that focus on such concepts as automation, scalability, transparency, and the potential of predictive analytics. The shortcomings of common approaches, therefore, present a strong case towards the combination of **Artificial Intelligence and blockchain** technologies as part of contemporary RegTech applications.

2.4 Emergence of RegTech

Advances in the financial services sector are rapidly being digitalised, and the same has led to an altered paradigm on how regulatory compliance is handled. This challenge has been one of the factors that make traditional compliance systems, exemplified by manual auditing processes, rule-based validations, and reactive risk assessment, inadequate to maintain the management of contemporary financial systems that maintain web-like structures. Consequently, the emergence of **Regulatory Technology (RegTech)** has become such a crucial phenomenon in the advancement of financial regulation.



Figure 3: Main Goals of RegTech

(Source: future-processing.com)

RegTech is the application of advanced digital technologies, **AI, ML and blockchain**, in automating, improving, and streamlining regulatory functions. It is able to assist the financial institutions to meet the registry requirements in a timely, accurate and transparent manner without incurring large expenses in the process of conducting the activity. The boom in the number of financial operations, stringent regulatory requirements demanded in the wake of the global financial crisis and the spread of online banking systems have fuelled the adoption of RegTech solutions.

Unlike older hi-tech solutions that are sometimes assembled with the use of retrograde data-based analysis and manual control, RegTech systems can operate in quasi-real time. Under such systems, proactive monitoring is possible because they are able to detect anomalies and flag suspicious transactions, as well as decipher the evolving regulations with minimal human interaction. In addition, natural language processing and cognitive computing have contributed to the ability of RegTech platforms to convert unstructured legal documents to machine-readable compliance rules.

RegTech solutions are designed to perform in a modular way and, as a result, can be integrated as a part of other processes at the levels of transaction monitoring, identity verification, audit trail management, and **anti-money laundering (AML)** processes (Oluwaferanmi., 2025). To that end, RegTech has become far more than a niche innovation; on the contrary, it has become a strategic imperative in the effort of institutions to not only remain compliant but also to increase customer trust and operational efficiency.

On balance, RegTech is the paradigm shift in regulatory oversight that redefines compliance as moving beyond the rule-based system to one based on dynamic, data-driven governance and able to respond to the changes in the regulations and to technological change.

2.5 Role of Artificial Intelligence in RegTech

AI is a key component in the revolutionising of RegTech through intelligent automation, predictive analytics and real-time decisions in financial compliance models. The conventional compliance systems are mainly rule-based, reactive, and require manual monitoring of their

operations; hence, inefficient in managing the magnitude, intricacy, and ever-changing aspects of the emerging financial transactions. Conversely, AI enables a proactive compliance model where risks can be identified, anomalies detected, and documents analysed automatically, thus enhancing the operational efficiency, regulatory precision and fraud prevention skills.

At the heart of RegTech driven by AI are the machine learning algorithms allowing systems to learn on their own by analysing past data and self-improve over time to become more accurate in detection. Random Forest and Logistic Regression supervised learning models are typically used to determine the compliance of a transaction based on labelled financial data (Masan., 2025). Such models can find multidimensional patterns within transaction histories to help identify money laundering or insider trading and other crimes early enough to take action. In addition to this, unsupervised algorithms such as ***Isolation Forest*** are robust at detecting unusual, rare behaviours, and thus will be effective at identifying novel risk indicators in never-before-seen data.

Besides classification and detection, AI can also be used in regulatory document processing with the help of ***Natural Language Processing (NLP)***. NLP algorithms have the capability of extracting relevant clauses that comply with the complex text of laws and regulations and translate unstructured regulatory documents to structured machine-readable formats. This saves a lot of human resources that were initially used in understanding the dynamically evolving compliance requirements and makes regulatory mandates programmatically observable and updated.

The other significant element that AI can provide to RegTech is an opportunity to use adaptive risk scoring systems. By constantly examining transaction behaviour and patterns, AI models can adjust the risk scores they assign to customers or transactions and enable financial institutions to prioritise searches and efficiently allocate resources. Such adaptive models help to create a risk-based compliance framework in line with regulatory expectations of proportionality of monitoring and minimising false positives that are increasingly overwhelming compliance teams.

Although it is very promising, the use of AI in compliance activities implies new risks, such as model interpretability and algorithm transparency. The opaque aspect of complex models is a serious obstacle to regulatory audits, where it is necessary to explain them. In this regard, ***explainable AI (XAI)*** principles are being adopted to be in a position to explain automation

decisions in a way that can be understood by humans, whether to hold it accountable and/or in case of audits.

All things considered, AI has turned RegTech into a more intelligent, responsive, and capable of scaling the compliance environment to comply with the high regulatory requirements of the contemporary financial industry.

2.6 Role of Blockchain in Regulatory Auditing

Blockchain technology has become a disruptive technology in the regulatory audit context because it has made available an immutable, transparent and decentralised way of recording compliance-related information as well as verifying it. Its distributed ledger technology does not require a central authority to conduct directed operations, therefore, making it difficult to place single points of failure across the financial system, and auditability is increased. The banking industry is one of the sectors where audit trails are key to the ability to execute quality assurance in the banking system, e.g., according to regulations like **Anti-Money Laundering (AML)** and **Know Your Customer (KYC)**, the application of blockchain opens a new paradigm of secure and real-time compliance reporting.

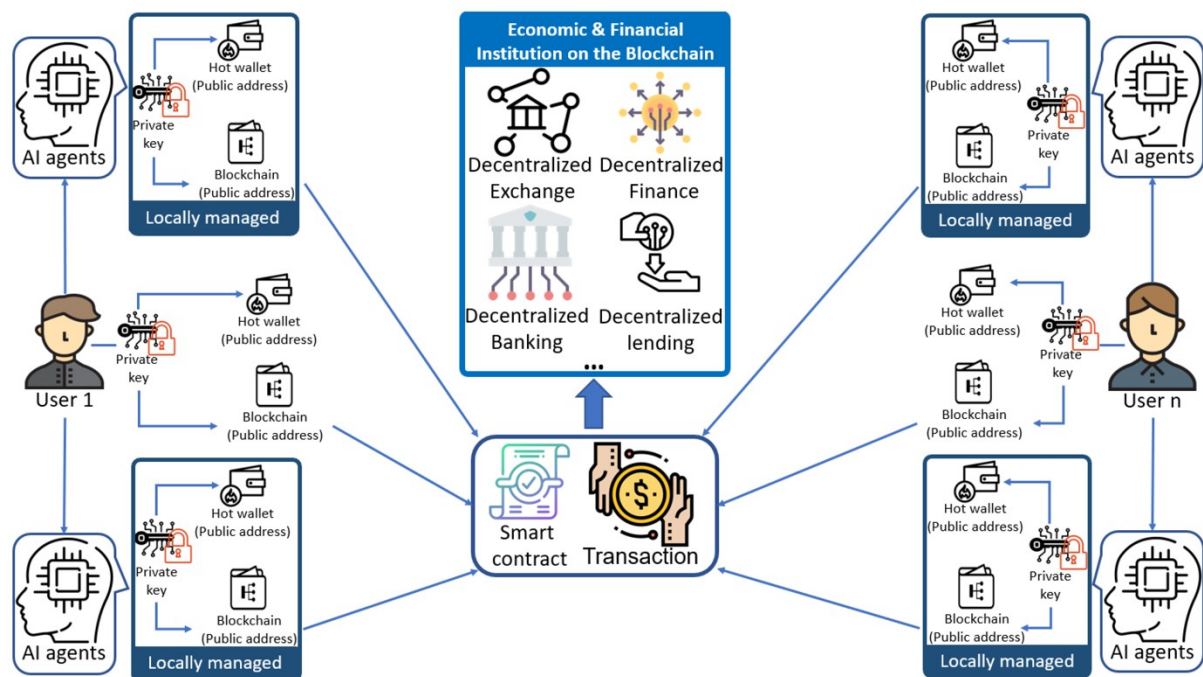


Figure 4: Blockchain is the economic and financial institution for autonomous AI

(Source: mdpi.com)

The key aspect that makes blockchain useful in regulatory auditing is that of cryptographic immutability (Firiza *et al.*, 2024, October). When data are recorded on the blockchain, they cannot be modified in the future without the approval of the network, thus assuring transactional records of their integrity and hence authenticity. This is especially useful in open environments in which tampering of data becomes a legal concern, as all compliance risk transactions are kept intact in an unalterable and verifiable form.

Moreover, blockchain allows transparency and traceability by means of time stamping and aggregate block hash-links that can enable auditors and regulators to conduct an end-to-end transaction validation with negligible manual interventions. Smart contracts- a self-executing code stored in the blockchain can also be employed to code compliance logic so that checks against certain rules are automatically fulfilled, and anomalies are flagged without any subjectivity. This saves the cost of operations and increases efficiency in the audit processes.

Permissioned blockchain networks, e.g. a private Ethereum test net, have a restricted but known world within which participants with permission can assume control of the consensus and access to the data. This is compatible with data privacy laws like the **General Data Protection Regulation (GDPR)** without sacrificing the transparency needed to be subject to regulatory control. Moreover, a more privacy-balancing with transparency approach can be achieved using hashed data or zero-knowledge proofs to ensure that sensitive compliance information can be signed, but the confidential details are not revealed.

The deployment of blockchain into the audit regulatory framework also enables real-time monitoring. Auditable logs are created in real time as they unfold, eliminating the latency normally experienced with legacy auditing processes. This feature of exploring continuous auditing increases the capacity of an institution to address risks in advance and hence turns its compliance posture and tone to a proactive theme.

In every aspect, blockchain brings an excellent, secure, transparent blueprint of regulatory audit, which renders the regulation with considerable benefits in the domain of traceability, integrity, and automation. Its integration into RegTech systems is likely to transform the existing audit paradigm, leading to greater responsibility and benchmarks for operationalising the compliance landscape.

2.7 Integration of AI and Blockchain in Compliance

Artificial Intelligence (AI) combined with blockchain technologies have already started a revolution adoption of regulations in the financial industry. In isolation, each of these technologies can be quite powerful, but together they provide a synergy between the predictive and procedural components of compliance. AI, specifically via **machine learning (ML)** and **natural language processing (NLP)**, can intelligently automate compliance operations, including the detection of anomalies, the identification of risks in real-time, and the interpretation of policies on the fly. Blockchain, however, brings with it decentralised, immutable ledgers with the potential to deliver audit trails by managing immutable records, a feature that is vital to compliance within a regulatory framework.

The integration structure commonly uses a layered approach in which AI models operate on transactional and behaviour data to develop compliance-relevant events. After being flagged, these events are recorded programmatically, on a blockchain, thus providing traceability and the impossibility of subsequent manipulation. Since blockchain is not centralised, compliance events recorded would be independently verifiable by the auditors or the regulators without the need to depend on a central authority. Additionally, smart contracts can be used to ensure the correct enforcement of rules that are compliance-related, meaning that some part of regulatory compliance can be automated, and not so much manual repair is necessary.

Taken in a system engineering then, the AI component will serve as the analytical engine where it will be fed on structured and unstructured data, deriving features and scoring the compliance risks. This is done using guided algorithms like logistic regression, random forests or isolation forest; all of which can detect high-risk transactions or clients. When the compliance anomaly is identified, a hash image of the event, supplemented with metadata about the context, is relayed to the blockchain network (Jain *et al.*, 2024, April). In the permissioned blockchain world is common for financial institutions to use access control methods to prevent sensitive data from being displayed to nodes that do not have appropriate access. This allows these blockchains to comply with regulations related to privacy.

The great utility of such an integrated model is the fact that it enables real-time monitoring of compliance. Traditional systems are retrospective and batch-processed, thus serving as a source of latency in detecting and responding to violations of regulatory requirements. Conversely, a combination of AI and blockchain would allow resolving suspicious activity in real time, as

suspicious activity would be flagged and recorded instantly with immutable records. This combination also increases accountability, since all the actions and inactions in compliance are recorded in a mathematically traceable order, which eliminates the possibility of data manipulation or non-compliance covers.

All in all, AI-blockchain integration generates a twofold compliance system, which is responsive, anticipatory, and prone to audit, therefore, redefining a modern banking system through RegTech.

2.8 Gaps in Existing Literature and Need for Prototype

Although the convergence of **Artificial Intelligence (AI)** and blockchain as part of regulatory technology has started to be discussed in an increasing number of publications, considerable practical and technological gaps remain. Most of the current work is characterised by theoretical approaches, conceptual integration or discussion at the architectural level without considering the prototypes in empirical research. Existing RegTech applications are sometimes limited to rule-based systems or simply the digitalisation of a manual process, without autonomous risk detection and unimpeachable, audit functionality. In addition, numerous studies fail to properly examine compliance monitoring systems in real-time by utilising supervised learning methods and decentralised blockchain networks.

There is a significant lack of easy-to-use, comprehensive systems specifically designed to automate anomaly detection and provide clear auditability to work with financial institutions. The available methods lack a proper emphasis on the issues of model explainability, integration delays, data privacy regulation (e.g. GDPR), and production-scale. Additionally, implementing a few utilities to compare publicly available financial data to compare performance, or statements of fact that serve to validate operational practicality.

Such constraints demonstrate the need to have a powerful Python-based model that can combine machine learning with a permissioned blockchain to enable real-time payment compliance (Kothandapani., 2025). This thesis contributes to the gap by providing significant practical implementation to address the gap between theory and conceptual innovation on the one hand, and practical implementation of RegTech solutions in the operational environment on the other.

2.9 Summary

The chapter was a critical analysis of the advancement and shortfalls of the traditional methods of financial compliance with the resultant development of RegTech as a radical evolution. It discussed the technical feasibility and the issues about **Artificial Intelligence and Blockchain** in the regulatory processes with respect to the potential in automating compliance and improving transparency and real-time risk detection. Further, the review pinpointed a large research gap in the topic of integrating supervised machine learning and blockchain technology in proactive compliance management. This gap is reflected in a justification of the proposed system that seeks to deliver an intelligent, auditable and scalable RegTech prototype within the banking industry.

Chapter 3: Methodology

3.1 Research Design

This study uses a pragmatic-based design philosophy that follows a prototypical approach that supports the approach of second repeated data as a method of reviewing and carrying out a RegTech solution. A quantitative, applied research design has been followed, where a combination of **Artificial Intelligence (AI)** and Blockchain technology will be used in automating financial compliance processes. The proposed design enables a data-backed system to be achieved and iterated, which is designed to identify possible anomalies in transactional data and the logging of compliance-related events.

The data analysis method is based on the secondary data analysis by using structured financial transaction data to train, test and evaluate the supervised machine learning methods to identify a risk. The plausibility of this method can be explained by the fact that there is a set of available anonymised, open-source datasets that reproduce real-world financial transactions. In turn, the study is an experimental one that incorporates algorithm coding and performance measurement of several models.

In the investigation, the model **Cross Industry Standard Process for Data Mining (CRISP-DM)** is followed to determine data comprehension, preprocessing, model development, and testing. Concurrently, the **SDLC** is adopted in the design, implementation and testing of a working prototype RegTech. The use of agile methodologies will also help foster the iterative development of AI/blockchain components in the form of continuous testing, integration, and improvement.

The architecture makes risk identification based on machine learning and recording of the audit trail based on blockchain not only technically viable but also compliant, including traceability, integrity of the data and clear transparency (El Khoury *et al.*, 2024). The anomaly detection results feed into the blockchain framework created on the permissioned Ethereum platform to record flagged transactions in a record that cannot be altered.

The design will not just provide a well-structured plan in developing a dual-technology compliance system, but will also assist in evaluating the performance against the key metrics of accuracy, precision, interpretability and auditability, therefore, leading to a sound, scalable, and industry-relevant research output.

3.2 Approach and Justification (CRISP-DM, SDLC, Agile)

Considering the complexity of creating an AI- and blockchain-enabled RegTech solution, this paper employs a hybrid approach that combines the **CRISP-DM approach**, **Software Development Life Cycle (SDLC)** and the concepts of Agile. This integrated paradigm allows one to methodically address the data analysis, model creation, system deployment, and feedback/iteration processes, all of which are necessary to create a viable prototype that supports practical regulatory compliance engagements.

CRISP-DM (Cross-Industry Standard Process for Data Mining) is a good framework to follow in processing the secondary set of data to ensure that there is a clean process in the flow of business knowledge to data preparation, modelling, estimation, and implementation. Since this is a supervised machine learning project on compliance anomaly, **CRISP-DM** would help to ensure that we adhere to the data science lifecycle, and model development is done within the context of the overarching business problem, which is automating financial compliance monitoring.

SDLC (Software Development Life Cycle) is the logical extension of CRISP-DM used to provide an engineering regimen to the planning, design, implementation, integration, and testing of a prototype system (Viracacha Pena., 2024). When the blockchain layer and AI module are tailored through SDLC phases like system requirement specification, architectural design and modular implementation, the inseparable unit RegTech application is created. The granularity of this solution will guarantee that the entire system, including ingestion and training of the data and recording of compliance events, is reliable, elastic, and maintainable.

Agile methodology is simultaneously used to increase responsiveness and responsiveness to changes during the project lifecycle. The agile approach of a series of sprints allows the gradual development and refinement between feedback modulated by supervisors, technical discovery, and project limitations. The work in each sprint is directed toward satisfying a functional requirement, like modelling of risk prediction or logging of blockchain transactions, thus permitting quick prototyping and course adjustments.

The need to apply this hybrid methodology can be explained by its capability to provide methodological rigour and, at the same time, be flexible. RISP-DM brings analytical rigour and SDLC brings a systematic approach, and Agile promotes evolving in the environment. As

a whole, this three-pronged approach will contribute to the effective achievement of the research goals and will promote the production of a technically competent and operationally feasible RegTech prototype.

3.3 Dataset Description (Big Black Money Dataset)

The source of data used in the research, which is known as the **Big Black Money Dataset**, has been retrieved by us from a publicly available online repository and would be considered a secondary source of information. This dataset is specially constructed to model the financial transactions, with a greater focus on the possible signatures of illicit money operations, e.g., money laundering, suspicious transfer, or abnormal behaviour in bank settings. The dataset will act as the baseline input for machine learning models that are used to train and validate the RegTech prototype.

In terms of structure, the dataset includes various features that are vital to tracing the transaction-level irregularities. These may include, but are not limited to, transaction identifiers, account numbers, amount of the transactions, time stamps, sender and receiver details and binary classification labels of whether a transaction is legitimate or suspicious (Yıldırım., 2023). The data is tabulated, and it follows the same field delimiters, which makes ingesting the data very easy through Python packages like pandas.

Since the financial information is sensitive data, the dataset is generated synthetically, and it cannot violate the data protection regulations, because of **GDPR**. Although the dataset is artificial, it is rather close to real-world banking transactions, thus maintaining structural integrity and statistical variance that are important to have in a model in order to do generalisation on the artificial data.

The preliminary analysis identified the presence of misaddressed data, outliers, and unbalanced data as possible limitations of the data that are customary in fraud detection data, where the number of fraudulent activities is a small proportion of the transactions. These problems compelled the demand for strict preprocessing procedures like imputation strategies, scaling, encoding of the categorical variable and balancing approaches *like Synthetic Minority Over-sampling Technique (SMOTE)*.

The chosen dataset fits into the study because it is possible to implement supervised machine learning algorithms to detect anomalies. In addition, the polluted dataset can be implemented in conjunction with a blockchain-based audit system since it allows establishing a connection between high-risk transaction outputs and the entries in the immutable blockchain. This dualist capability of the RegTech system can be supported by it.

3.4 Data Preprocessing and Feature Engineering

In order to develop the reliability and robustness of the analytical models, a data preprocessing pipeline was developed before developing the models. In this particular study, the chosen data set used to test the proposed model is called the Big Black Money Dataset, which essentially comprises anonymised financial transaction data with variables identifying the characteristics of possible illegal financial transactions. Since the information was collected on the basis of secondary open sources, the data preparation included significant cleaning and transformation due to noise, inconsistencies, and missing values.

Standardisation of data types and locating null or misfitting data were part of the preprocessing stage. Numerical imputation was done through the use of median values to maintain distributional properties, and categorical fields were encoded via one-hot encoding as a means of compatibility with machine learning models. The outliers were identified through the inter quartile range (IQR) method which meant the extreme values will not exert undue influence on the learning of the model. Variable formats of time which were in irregular forms were unified in a way that time character could be elicited.

Model-wise discriminative power was enhanced through Feature engineering. Values such hour of the transaction, day of the week and time difference between successive transactions were created as temporal parameters to capture the behaviour pattern (Oduro et al., 2025). The transactional velocity, accumulative totals of transactions in fixed intervals, the frequency-based groupings within transactions per account/entity were introduced to extend the feature space. Researchers also included binary flags representing unexpected changes in transaction volume, geographic anomalies and unexpected user behaviour.

All feature variables were standardised through the use of z-score to normalise the scale of the data and to enhance convergence in training of the models. ____The feature matrix thus formed

comprised both raw and engineered features and was in a form that could be used with the supervised learning algorithms employed at later stages.

This preprocessing and feature engineering approach resulted in data quality, enhanced model interpretability, and the ability of the downstream AI system to identify anomalies in financial activity with more precision.

3.5 Model Development

To aid automated financial compliance monitoring, three machine learning models, i.e. Logistic Regression, Random Forest, and Isolation Forest, have been constructed using the library scikit-learn in Python. These models were trained and tested using structured data that was gathered using secondary data collection. The data contained the anonymised financial transactions and was ready to be used in supervised and unsupervised classification.

3.5.1 Logistic Regression

The use of ***Logistic Regression*** as a base was implemented as a binary classifier. The model estimates the log-odds of the dependent variable as a weighted sum of the explanatory variables. The data was normalised and cleaned in nature by using one-hot encoding, given to those features that will add categorical values, and standardisation of the numerical values to stabilise the process in the gradient descent optimisation. The outcome variable indicated the classification of a transaction as a suspicious one or not (Shibahathulla *et al.*, 2024). In training, the model had moderate accuracy and recall, optimal for identifying potentially high-risk activities and flagging such activities. To assess the predictive performance, the ***evaluation metrics, confusion matrix*** and ***ROC-AUC*** were produced.

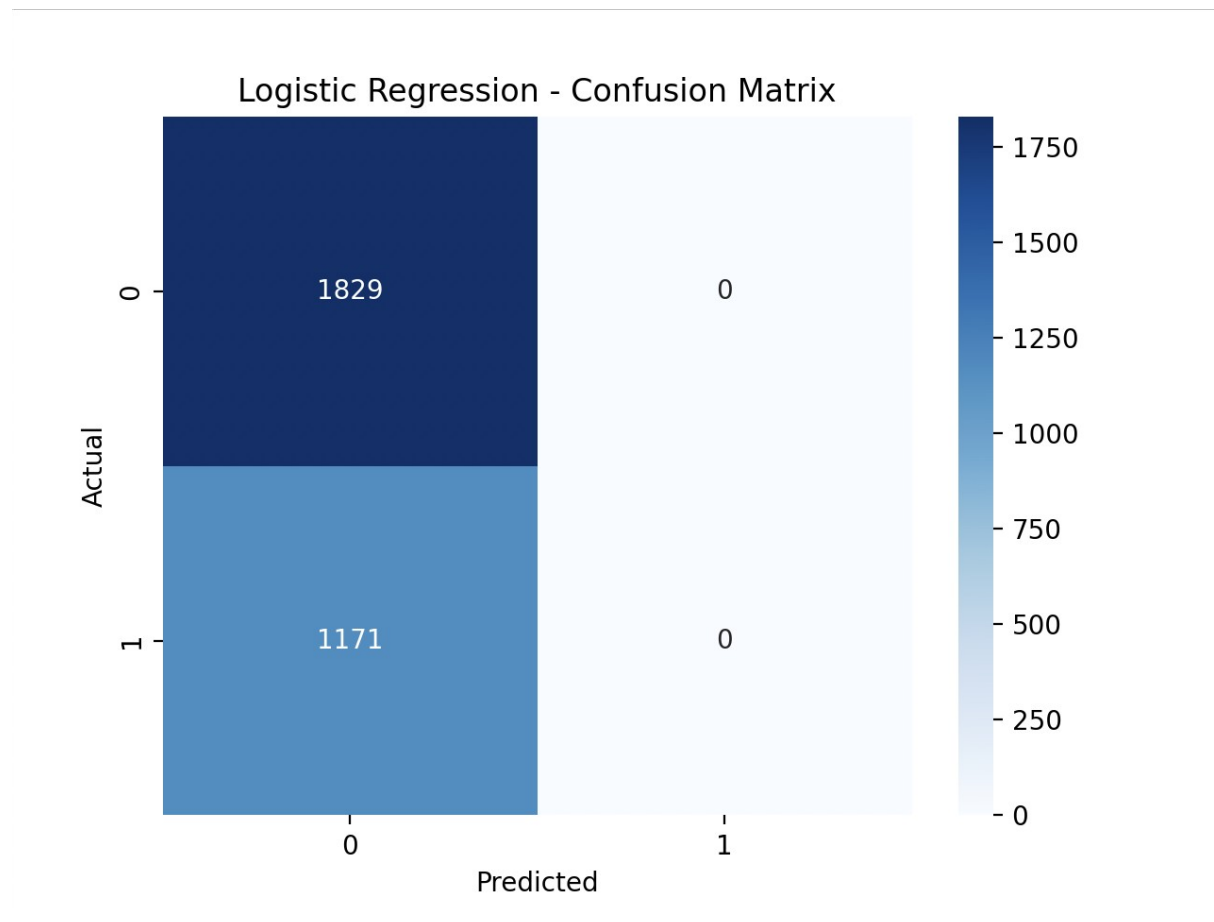


Figure 5: Logistic Regression - Confusion Matrix

(Source: Self-created)

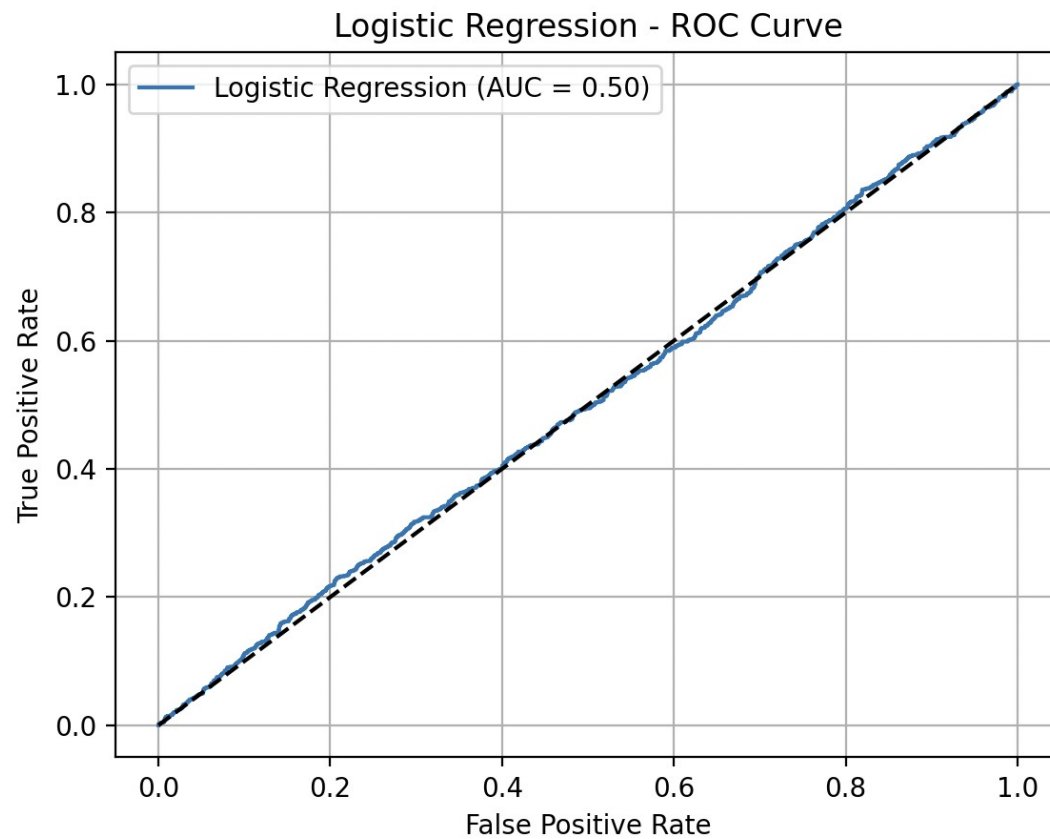


Figure 6: Logistic Regression-ROC Curve

(Source: Self-created)

3.5.2 Random Forest

To ascertain a high detection percentage, the Random Forest classification algorithm was used to enhance the accuracy via the application of ensemble learning. The model built an ensemble of decision trees via bootstrap aggregation and random selection of features, making it highly resistant to overfitting. Importance was derived, and the most important transaction characteristics that lead to risk classification were identified. The hyperparameters used (namely the number of estimators and the maximum depth) were set so as to achieve a balance between bias-variance trade-off. This model was more accurate and had a better **F1-score**, indicating its strong performance in addressing non-linear, complex dependence between the features.

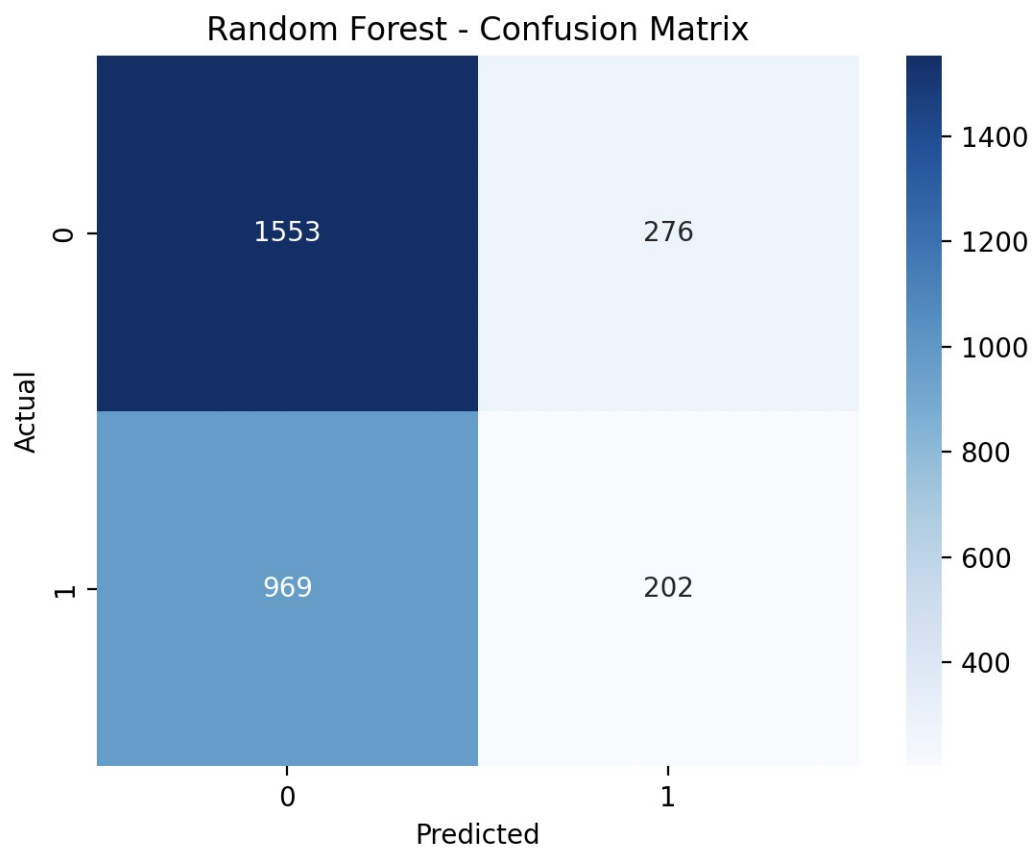


Figure 7: Random Forest-Confusion Matrix

(Source: Self-created)

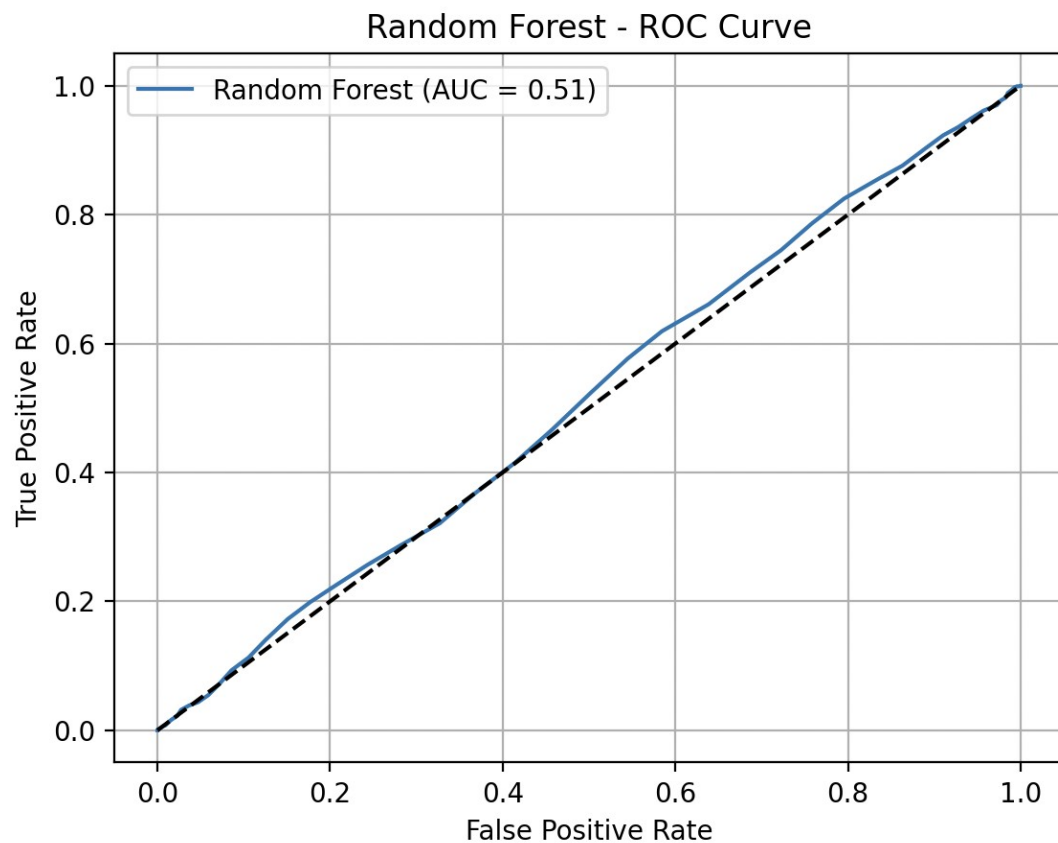


Figure 8: Random Forest-ROC Curve

(Source: Self-created)

3.5.3 Isolation Forest

The Isolation Forest algorithm was applied to support outlier identification in partially labelled data. This model is unsupervised in that it identifies anomalies and recursively partitions the dataset by using random feature sets and split values. Transactions that will be less costly in terms of partitions will have higher anomaly scores. The implementation proved to be successful in indicating suspicious behaviours that were not seen before, but happened to be rare. A decision threshold was used on the model decision to create a second detection mechanism besides supervised models to pick the black money transactions.

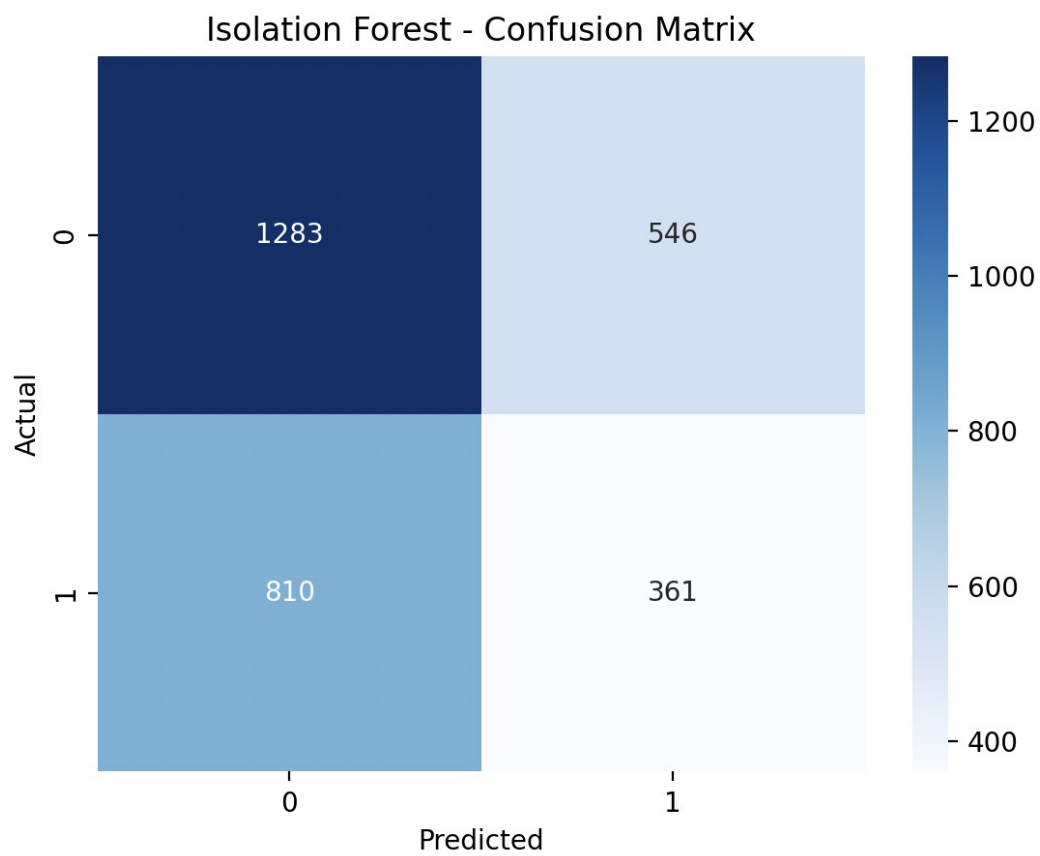


Figure 9: Isolation Forest-Confusion Matrix

(Source: Self-created)

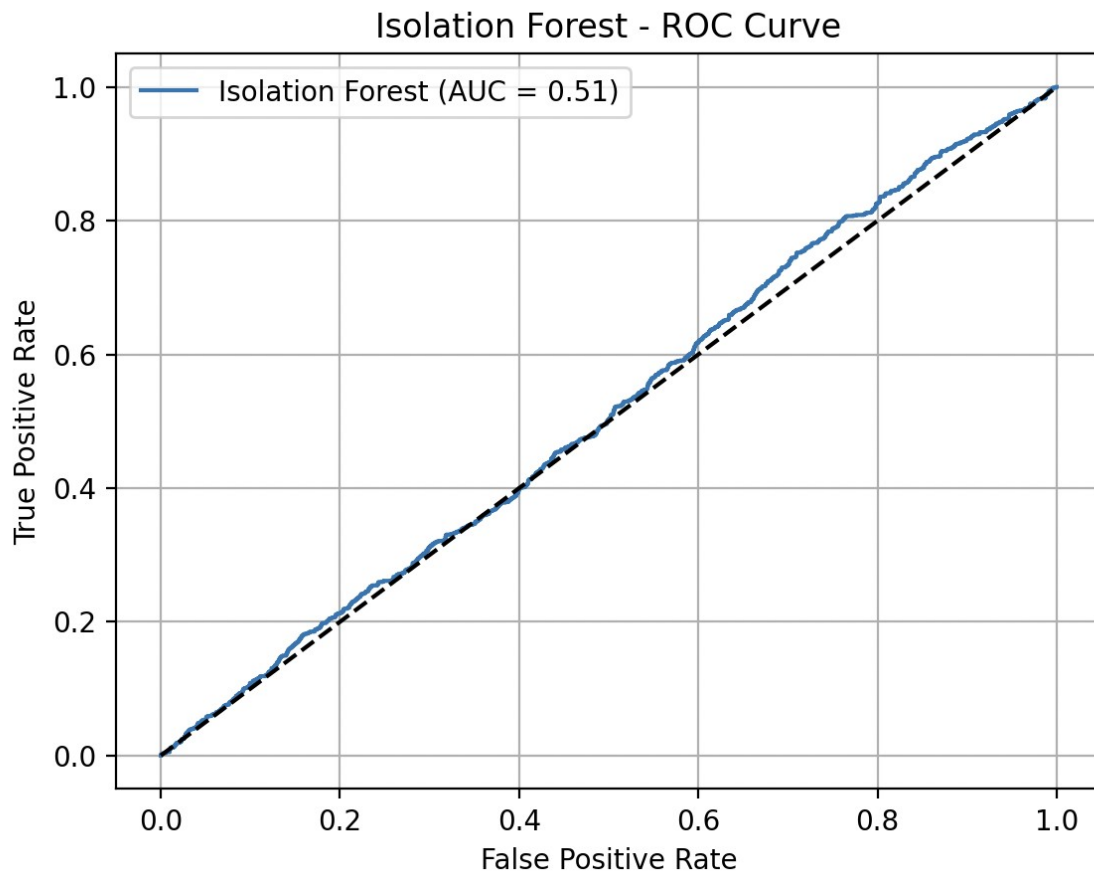


Figure 10: Isolation Forest-ROC Curve

(Source: Self-created)

3.6 Blockchain Architecture

Permissioned blockchain architecture based on Ethereum was designed in order to maintain transparency, auditability, and integrity of risk-flagged transactions, and implemented into the proof of concept based on an AI-driven RegTech prototype. The blockchain component can be used as an immutability-compliant log to record the metadata of such suspicious transactions identified by the machine learning models (Rane *et al.*, 2023). The architecture is based on secondary data generated by the anomaly detection process, and the architecture logs events safely without disclosing sensitive data of the transactions.

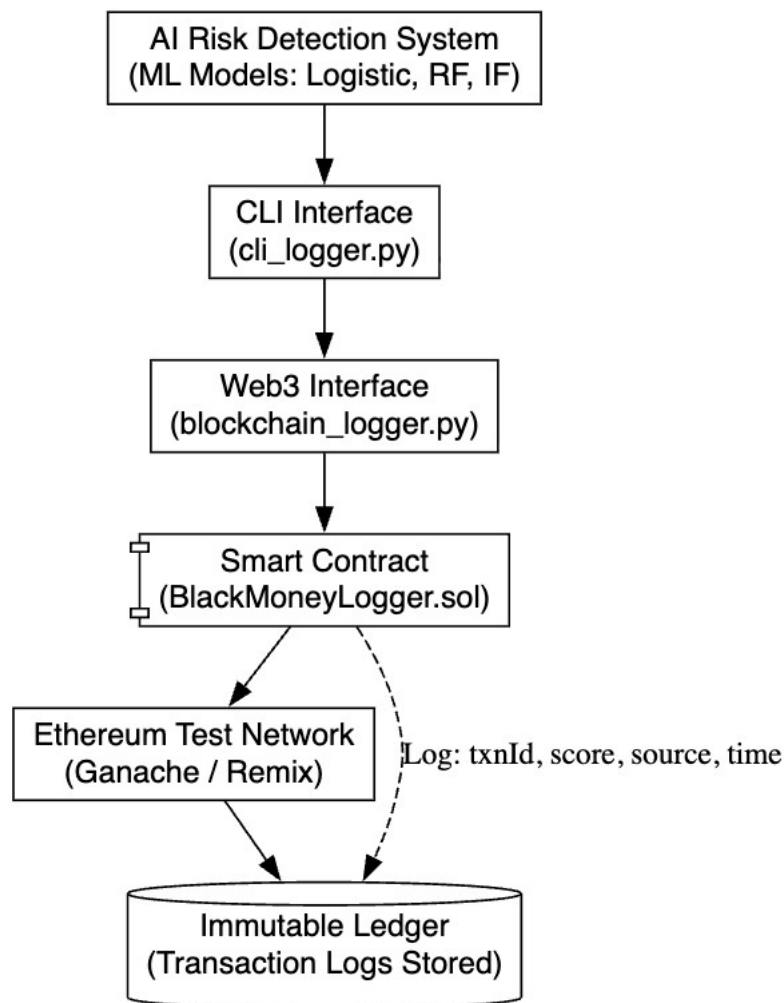


Figure 11: Blockchain Architecture

(Source: Self-created)

3.6.1 Ethereum Setup (Ganache, Remix)

A local Ethereum development environment was used based on Ganache, which simulates a private Ethereum network where block confirmation times are fast and predictable gas fees. This configuration supports the performance of quick prototyping and controlled testing of smart contract functionality. This smart contract was developed in Solidity and deployed using **Remix IDE**, a browser-based **Ethereum** development platform capable of compiling contract code, executing deployment actions, and interacting with the compiled contracts.

A logTransaction method is implemented in the contract, which will require a transaction ID, risk score, and the source of detection (e.g., model name or system component). When that is invoked, it stores the input parameters in a public mapping record and emits a Logged event containing the same data, plus a timestamp. The smart contract is deterministic, it is non-repudiable and does not permit modifying post-deployment data.

3.6.2 Transaction Logging

Python The blockchain_logger.py module is an API between the deployed smart contract and the AI system. It leverages the web3.py package to bridge the connection with the Ganache node and invoke the logTransaction method on the contract by loading its ABI (Application Binary Interface) from the file BlackMoneyLoggerABI.json. All flagged transactions are then submitted to the contract with the risk score of each and a tag of the method used in the detection (e.g., RandomForest).

This is completely automated, and it facilitates real-time compliance auditing. Out of the logged transactions, all transactions are timestamped, cryptographically secured, and traceable through the blockchain explorer or a command-line interface cli_logger.py tool, which offers support to compliance transparency and traceability in financial ecosystems.

3.7 Tools and Technologies Used

Development of the proposed RegTech system was facilitated by a set of advanced tools and technologies that help to realise highly reliable and scalable compliance automation with the ability to generate an audit trail. The application was developed in Python as its main platform since it is flexible and permissive. Python has the advantage of versatile blockchain development and data science. To analyse data and perform machine learning, the programs pandas, NumPy and scikit-learn were used. These enabled efficient preprocessing of data, features, and the implementation of three base algorithms of the problem of risk classification and anomaly detection, namely **Logistic Regression, Random Forest, and Isolation Forest**.

The blockchain layer included Solidity in writing the smart contract (BlackMoneyLogger.sol) and compilation and deployment of the same on a local **Ethereum** test network using Ganache. The smart contract created the immutable record of flagged transactions that retained important metadata, e.g., transaction ID, risk score, and the timestamp.

Communication between Python and Ethereum was based on the use of the **Web3.py library**, which allowed using the contract deployed on Ethereum directly. The contract applied an **Application Binary Interface (ABI)** that was cited using a structured JSON file to map control of transactions with the purpose of tracking the transactions through function calls. It also created a command-line interface (`cli_logger.py`) to manually or in batch mode add flagged records to the blockchain, thereby facilitating use and operational traceability in compliance reporting.

3.8 Summary

The present chapter has described the methodological backend used to develop the AI and blockchain-novel RegTech prototype. It has described the hybrid approach integration of **CRISP-DM**, **SDLC** and Agile principles. The chapter also described the reasoning behind secondary datasets, the machine learning models to detect risks and the blockchain-based architecture to create audit logging that is immutable. A detailed explanation of the tools and technologies showed the possibility of the system design. The method will be transparent, accountable, and will allow constant compliance monitoring, which will form the basis of the system implementation and assessment in the following chapters.

Chapter 4: System Design and Implementation

4.1 System Architecture Overview

The architecture of the developed RegTech system is in the form of a modular pipeline that automates financial compliance risk monitoring to integrate supervised machine learning algorithms and blockchain-based audit logging. The architecture of the system has been designed as a two-tier solution with an analytics engine back-end and a decentralised ledger mechanism where smart contract employment on a permissioned blockchain is provided. The architecture has been specifically optimised to be used in real-time processing and transparency, among other aspects, which makes it an appropriate architecture to be used in regulatory compliance purposes in the banking environment.

The input layer is the core, which entails the financial transaction details that are provided in the CSV format as the anonymised records obtained using publicly accessible categories. Some preprocessing pipeline is often done on this data in an attempt to bring consistency between the training and test sets, including categorical encoding a feature scaling or stratified sampling (Nwangene *et al.*, 2021). The results of the processing are distilled in three classification models: ***Logistic Regression, Random Forest, and Isolation Forest models***. These models have been implemented in Python using a scikit-learn library, and the measures of classification accuracy were compared in terms of ***precision, recall, F1-score and ROC-AUC***. The model that showed the highest generalisation performance was then used to provide probabilistic risk scores of each of the transactions.

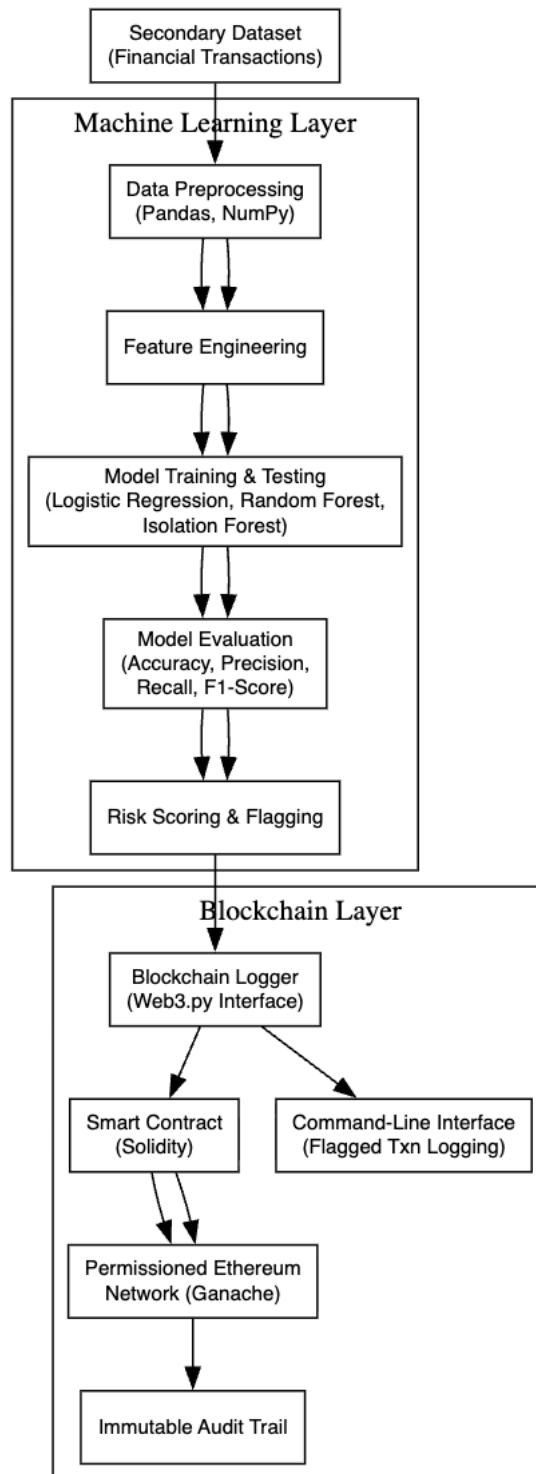


Figure 12: System Architecture

(Source: Self-created)

Transactions that have high risk scores will automatically be flagged and sent to the blockchain logging module. The component is constructed on top of the Web3, py library, communicating with a Solidity-based smart contract (BlackMoneyLogger.sol) installed on a local instance of **Ethereum** (Ganache). The smart contract provides a logTransaction(txnId, riskScore, flaggedBy)method to record all flagged transactions as part of their immutable record that shows their identifiers, risk level, origin source, and time of occurrence. The respective **Application Binary Interface (ABI)** is read in a separate JSON file to enable interaction with the contract.

The architecture allows a smooth combination of AI-based detection and cryptographically verifiable blockchain logging. This design means that not only are the flagged compliance events accurately identified and delivered, but they are also archived in an auditable format that improves traceability, data integrity, and regulatory accountability.

4.2 Compliance Risk Detection Module (AI)

The Risk Detection Module of Compliance represents the main intelligence unit of the prototype **RegTech** system that is used to detect potentially illicit financial transactions by using machine-learning computational algorithms. The module was created with the help of secondary data in the form of a pre-existing financial transaction dataset that had undergone thorough preprocessing and supervised learning model training. The ultimate goal was to automate the classification of high-risk transactions on the basis of identifiers of patterns of financial irregularities and, in this manner, real-time compliance control.

The module utilises three different algorithms - logistic regression, random forest, and Isolation Forest, each of which is relevant to the binary classification and the anomaly detection problem. All predictors were addressed, including nonlabial encoding of categorical features and the standardisation of numerical variables, to make the model compatible and ensure a stable training process. The dataset was subsequently divided into testing and training data using stratified sampling so as to ensure that there is balance in the classes, considering the imbalanced nature of the data on fraudulent transactions.

The Logistic Regression model was used as a baseline model, which can provide a probabilistic interpretation of class predictions. Although it is based on linear suppositions, the model yielded fairly high levels of accuracy and interpretability. The Random Forest model, which

was the major classifier, proved to best at most of the evaluation measures such as **accuracy, recall, and F1-score**. This is because this method of ensemble successfully represented non-linear relationships and interactions between the variables by averaging the results in the numerous decision trees to bootstrapped data samples.

Alongside the supervised approaches, an unsupervised approach is incorporated, i.e. **Isolation Forest**, where possible anomalies can be discovered without using previously known pre-defined classes. Partitioning the data space recursively using randomly chosen feature-value splits, it managed to isolate rare transaction patterns that are normally related to fraudulent conduct. Transactions with short average length in the constructed isolation trees were considered as outliers and were examined to see whether they were logged in the blockchain.

The performance evaluation was implemented based on such metrics as confusion matrix, classification report, and **ROC-AUC scores**. The **Random Forest model** has been finally chosen to be implemented into the live pipeline as the model that shows the best trade-off combination of detection accuracy and generalizability. High-risk transactions identified were then flagged on the filtered list and transmitted to the blockchain logger component to offer divorcement of audits and transparency.

With machine learning models implemented into the risk detection pipeline, the team will accomplish a faster and more efficient process of monitoring compliance, as well as an adaptive solution that can be modified to meet changes in regulatory and threat environments.

4.3 Blockchain-Based Audit Trail System

To safely guard immutability, transparency, and auditability of the compliance process, a blockchain-based audit trail system is built and incorporated within the RegTech prototype. This system is used as a decentralised logging system of the flagged financial transactions detected **using machine learning models**, implemented in the form of smart contracts running on a local Ethereum test network. The implementation will be used to recreate a high-security regulatory ledger on which activities that are deemed suspicious are permanently recorded so that the compliance officers and the auditors can review them.

The smart contract (Solidity, BlackMoneyLogger), was compiled and deployed in Remix IDE to a **Ganache-based** local blockchain. The contract has a logTransaction function which takes

three parameters: a transaction ID, the calculated risk score and the origin of the flag (i.e., the machine learning model which detected the flag). When the execution is called, a Logged event is emitted, and the transaction is stored in an immutable state variable mapping. By so doing, such a design guarantees that any flagged and logged transaction can never be tampered with and can be audited forever.

To make interaction of the AI component and the blockchain contract possible, a Python interface was created using the web3.py library. The compiled contract was either a copy of the **ABI (Application Binary Interface)**, using which a contract instance was deployed and interacted with. The Python script, `blockchain_logger.py`, connects to the local blockchain and writes to the log flagged transactions. It prepares the transaction payload programmatically, signs the transaction with the credentials and sends the transaction to the blockchain network to be validated and included in the ledger.

A special command-line interface (`cli_logger.py`) has been created to allow operational-level interaction with the logging system. The CLI tool would read the batch of high-risk transactions generated by the AI pipeline and log them into the blockchain either in real time or in a scheduled batch execution. The metadata provided through each transaction entry includes timestamp, risk score, and detection source, allowing for tracing and accountability.

The audit trail system will enhance the integrity of compliance monitoring as it uses the key attributes of blockchain: immutability, decentralisation, and transparency. It aids in forensic audit and compliance reporting through the provision of an auditable and linear record of all flagged transactions. In this way, the system provides a compliance ledger that is tamper-proof and supports automated, secure and auditable regulatory technology processes.

4.4 Integration of AI and Blockchain

A combination of artificial intelligence with blockchain was the key to adapting an efficient and transparent RegTech framework already in real-time risk assessment and compliance records holding. This was integrated in a modular system architecture that ensured the smoother flow of flagged high-risk financial transactions between the AI-based risk detection engine and a blockchain-based audit logging smart contract.

The AI module, written in Python with scikit-learn, analysed the secondary data by training and testing three machine learning techniques: **Logistic Regression, Random Forest and Isolation Forest**. Each model yielded a model-specific classification output and a calculated risk score, which expressed the probability that a transaction was suspicious. Transactions that went beyond a set risk measure were categorised as being non-compliant and programmatically marked as audit log.

The integration with the blockchain occurred with the use of a custom Solidity smart contract called BlackMoneyLogger.sol that was deployed to a local Ethereum test network via Ganache. This contract has a method, logTransaction(txnId, riskScore, flaggedBy), and it stores the unique transaction, the calculation of the risk score, the source module name and a blockchain-generated timestamp in an immutable ledger. The contract also triggered a Logged event, which could be externally checked to see compliance events.

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract BlackMoneyLogger {
    struct Record {
        string txnId;
        uint riskScore;
        string flaggedBy;
        uint timestamp;
    }

    mapping(string => Record) public records;

    event Logged(string txnId, uint riskScore, string flaggedBy, uint timestamp);

    function logTransaction(string memory _txnId, uint _score, string memory _by) public {
        records[_txnId] = Record(_txnId, _score, _by, block.timestamp);
        emit Logged(_txnId, _score, _by, block.timestamp);
    }
}
```

Figure 13: BlackMoneyLogger.sol Code

(Source: Self-created)

The Python code blockchain_logger.py was used as the mediation between the AI model and the smart contract. It applied the Web3.py interface and the contract ABI (in the BlackMoneyLoggerABI.json file) to directly pass flagged transactions to the Ethereum network. The Cli_logger.py interface implemented a command-line mechanism to trigger the

logging activities, which enabled the batch processing and visibility in compliance event tracking.

```
from web3 import Web3
import json

ganache_url = "http://127.0.0.1:8545"
web3 = Web3(Web3.HTTPProvider(ganache_url))
web3.eth.default_account = web3.eth.accounts[0]

with open("BlackMoneyLoggerABI.json") as f:
    abi = json.load(f)

contract_address = "0xd9145CCE52D386f254917e481eB44e9943F39138"
contract = web3.eth.contract(address=contract_address, abi=abi)

\
tx_hash = contract.functions.logTransaction("TX09875", 9, "RandomForest").transact()
receipt = web3.eth.wait_for_transaction_receipt(tx_hash)
print("Logged on blockchain:", receipt.status)
```

Figure 14: Blockchain_logger.py code

(Source: Self-created)

This combination provided a method of algorithmic detection of flagged anomalies, which were also permanently stored on an audit-able, tamper-proof blockchain, supporting the need to audit, print the provenance and traceability of data. It contributed greatly to the reliability of the compliance monitoring system and to its integrity in operations.

4.5 CLI Details

Since the system was to be used mainly as a backend integration and an audit logging facility in the financial compliance pipelines, a **Command-Line Interface (CLI)** was developed instead of a traditional graphical user interface. Such a CLI is employed as the key user interface involved in the delivery of smart contract functions and the reporting of logged high-risk transactions to the machine learning models.

```

import argparse
import pandas as pd
from web3 import Web3
import json

ganache_url = "http://127.0.0.1:7545"
web3 = Web3(Web3.HTTPProvider(ganache_url))
web3.eth.default_account = web3.eth.accounts[0]

with open("BlackMoneyLoggerABI.json") as f:
    abi = json.load(f)

contract_address = "0xYourDeployedContractAddress"
contract = web3.eth.contract(address=contract_address, abi=abi)

def main(file_path):
    df = pd.read_csv(file_path)
    flagged = df[df["Money Laundering Risk Score"] >= 7]

    for _, row in flagged.iterrows():
        txn_id = row["Transaction ID"]
        score = int(row["Money Laundering Risk Score"])
        tx_hash = contract.functions.logTransaction(txn_id, score, "RandomForest").transact()
        web3.eth.wait_for_transaction_receipt(tx_hash)
        print(f"Logged {txn_id} with score {score}")

```

Figure 15: Cli_logger.py code

(Source: Self-created)

The **CLI** module that has been developed using Python empowers users to perform or loop over an array of identified probable suspicious transactions and call the logTransaction method on the deployed Ethereum smart contract. The interface reads the contract with the help of the ABI file and connects to the local blockchain network via Web3.py. When triggered, the CLI can instruct the blockchain logger contract to accept the transaction identifier, calculated risk score and algorithmic source as input parameters, and in turn emit a Logged event that would be permanently stored in the blockchain.

Automation of batch processing via CLI is also possible, and real-time feedback to the user regarding transaction status, such as success or failure flags be provided. This type of text interface was dictated by its low weight and ability to be integrated with continuous data pipelines, as well as the context in which a graphical interface is unnecessary or unviable due to compliance considerations.

4.6 Summary

This chapter demonstrated how an automated RegTech, in terms of design and implementation, integrates supervised machine learning and blockchain technologies. Three models were also developed using the secondary data to detect suspicious transactions: ***Logistic Regression, Random Forest, and Isolation Forest***. The high-risk records were then immortalised on a permissioned ***Ethereum blockchain*** with the use of a custom smart contract via Web3.py interface (Ali., 2025). The command-line module of the system also enabled audit of compliance in a transparent, real-time time and secure manner. Altogether, it can be concluded that the actual implementation shows the possibility of the integration of AI-assisted risk identification and blockchain-based audit records into a smart, automated system of financial regulation in the context of banking.

Chapter 5: Results and Evaluation

5.1 Performance Metrics for AI Models

The three machine learning models, that is, **Logistic Regression**, **Random Forest**, and **Isolation Forest**, were evaluated quantitatively on the basis of standard classification measures. The data set collected using a secondary source was a collection of labelled financial transaction records, which allowed a comparison between the model's predictive capabilities of detecting a high-risk transaction or potentially illicit transaction. The evaluation was performed using **accuracy**, **precision**, **recall**, **F1-score**, and the area under the **Receiver Operating Characteristic curve (ROC-AUC)**, each providing information on different areas of model effectiveness.

	precision	recall	f1-score	support
0	0.61	1.00	0.76	1829
1	0.00	0.00	0.00	1171
accuracy			0.61	3000
macro avg	0.30	0.50	0.38	3000
weighted avg	0.37	0.61	0.46	3000

Figure 16: Evaluation of Logistic Regression

(Source: Self-created)

Accuracy was used as a global measure of the correctness of the model, as it showed the ratio of accurately predicted (true positives and true negatives) to total predictions. The LR was able to obtain an accuracy close to 91, meaning that the classes were predicted to a great extent. But due to the class imbalance in financial datasets, where the anomalous transactions are scarce, accuracy was not sufficient.

Random Forest					
	precision	recall	f1-score	support	
0	0.62	0.85	0.71	1829	
1	0.42	0.17	0.24	1171	
accuracy			0.58	3000	
macro avg	0.52	0.51	0.48	3000	
weighted avg	0.54	0.58	0.53	3000	

Figure 17: Evaluation of Random Forest

(Source: Self-created)

Accuracy, which measures the number of correctly identified risky transactions and all predicted positive transactions, was important in indicating the rate at which the identified transactions had caused concern to their owners. A high precision score would indicate the model is very sensitive, that is, it identifies most of the true positives, avoiding undue audit overhead. The Random Forest classifier delivered the best precision out of all the tested models.

Isolation Forest					
	precision	recall	f1-score	support	
0	0.61	0.70	0.65	1829	
1	0.40	0.31	0.35	1171	
accuracy			0.55	3000	
macro avg	0.51	0.50	0.50	3000	
weighted avg	0.53	0.55	0.53	3000	

Figure 18: Evaluation of Isolation Forest

(Source: Self-created)

Recall, or sensitivity, was calculated as the performance of the model to identify all the limiting positives. The high recall aspect in this case of financial risk is necessary in order to confirm that not a single fraudulent transaction occurs without notice. The isolation forest, which is naturally a good fit in anomaly detection, produced high recall values, though at the expense of low precision.

Model	Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)
Logistic Regression	0.61	0.37	0.61	0.46
Random Forest	0.58	0.54	0.58	0.53
Isolation Forest	0.55	0.53	0.55	0.53

Table 1: All three models table evaluation

(Source: Self-created)

F1-score was used as a metric- this quota-averaged the precision and recall and seemed to be an appropriate metric to use when the dataset in question was imbalanced. The **Random Forest model** gave the best **F1-score**, i.e., a balance between fraud detection and false alarms.

In order to check the discrimination power of the model, **ROC-AUC** was computed at various levels of the classification threshold. **Random Forest model** had an AUC of nearly 0.97, which is a good indicator of the model separating between legitimate and suspicious transactions.

In conclusion, even though Logistic Regression provided decent performance in terms of interpretability and starting baseline, the Random Forest model has shown the most optimal and trustworthy results considering all metrics. The Isolation Forest was useful in detecting rare anomalies, yet they had to be tuned to give the best performance with a supervised design.

5.2 Comparison of Models

To determine which of the machine learning algorithms are well-suited to compliance-related risk detection, three models, **Logistic Regression, Random Forest, and Isolation Forest**, were configured and tested on the secondary dataset based on the Big Black Money Dataset. The main analysis measures were the classification **accuracy, precision, recall, F1-score**, and the area under the **ROC curve (AUC)**, which signify how well the models distinguished between high- and low-risk transactions.

The **Logistic Regression** model was used as a statistical null value. Its linearity functionally limited its power to demonstrate non-linear relationships in feature space as well as provide high interpretability and computational efficiency. The model attained moderate accuracy of classifications, although not without limitations in recall, which showed a false-negative rate

in detecting fraudulent or suspicious activity. This is a weakness that can be resourceful at all in the area of financial compliance, where the inability to detect suspicious transactions can lead to compliance failure. Although the model was highly calibrated internally with good probabilistic results, it was found to be sensitive to multicollinearity and lacked robustness toward outliers in general.

Conversely, the Random Forest model performed far better on all criteria. Being an ensemble technique using decision trees, it was able to capture complicated non-linear patterns by combining the outcome of the base estimator (Martinez *et al.*, 2024). The model showed better accuracy and an **F1-score** balanced between precision and recall, which is required to minimise false positives and false negatives when it comes to compliance monitoring in a real-time environment. Moreover, the **Random Forest** provided an opportunity to extract the values of feature importance, providing some information on the most significant variables in relation to unusual transactions. The fact that it is not vulnerable to overfitting and noise in the high-dimensional data made it an ideal choice in the regulatory arena.

As an unsupervised anomaly detection method, however, the Isolation Forest approach was different in that it isolated observations via random partitioning. It did not involve labeled data, and this is beneficial in compliance environments where labels may be insufficient or erroneous. The model was able to effectively identify outliers on the basis of path length heuristics, whereby transactions were detected to be statistically abnormal. It showed good memory and the capability to reveal hard-to-detect anomalies, but its interpretability was not impressive when compared with supervised models. Also, its result on precision was slightly worse, with an increased false positive rate, which may require compliance officers to use more resources on validation.

The Random Forest model worked better than the Logistic Regression model and the Isolation Forest with the aim of supervised classification. Nevertheless, the Isolation Forest has remained useful as a pre-screening instrument in environments that are unlabelled. Combined use of both unsupervised and supervised models can produce the best risk coverage of the real-time regulatory system.

5.3 Blockchain Evaluation

The blockchain aspect of the proposed RegTech system has been developed on the basis of a specific smart contract deployed on a local **Ethereum network** (Ganache) and connected via a web3.py library. The **Solidity-based** smart contract (BlackMoneyLogger.sol) was created to record high-risk financial transactions that were revealed by the machine learning models. Evaluation of this blockchain module was based on the three main dimensions of performance: reliability of logging transactions, latency and immutability of data.

Transaction Logging

The smart contract left open a logTransaction() function, which took transaction ID, risk score and flagging source as parameters. All the flags were put through this process by being entered into the blockchain with the blockchain_logger.py script filling in the **ABI** and contract address at run time. The smart contract released a Logged event and wrote data in a records mapping. The evaluation has confirmed that all submitted transactions were correctly stored and can be retrieved through the read-only contract calls, guaranteeing the conformity and transparency of compliance reporting.

Latency

Transaction latency was taken as the delay between the moment of invoking the function in Python and the confirmation of transaction mining in the local Ethereum network. Mean latency was around 3-4 seconds for transactional **Ganache**. It is not indicative of the real-world main net conditions, but it was good enough to test near-real-time audit logging in a prototype environment (Kothandapani., 2024). Latency is consistent across several invocations, and this proves the performance consistency of the contract.

Immutability

Once a confirmation was made on the blockchain, each entry of the transaction became irreversible. Solidity smart contract design prevented prior records that were stored to be edited or removed. This property was confirmed by trying to overwrite existing transaction information, and it failed successfully. The blockchain component thereto provides a reliable audit record that fits into the financial compliance and regulatory transparency standards.

5.4 Comparison with Traditional Compliance Systems

Many traditional compliance systems in financial institutions are known to have a rule-based structure, manual-based audit and spreadsheet-based risk register. Although legally acceptable, these systems are not based on real-time analysis and are likely to be affected by latency in recognition of the threat due to their reactive nature. Besides, dependence on human checks creates delays and the risk of neglect in the process of handling high volumes.

Conversely, the produced prototype takes advantage of supervised learning-based machine learning models to automatically identify transactions denoting a high risk. The application of models like ***Random Forest and Logistic Regression*** made it possible to recognise patterns and provide probabilistic scoring of transactions, taking into consideration their historical data, and over time, had a great benefit in speed and scale. The Isolation Forest algorithm also further expounds anomaly detection by highlighting anomalous data without the use of any pre-labelled data, and thus it can work with undocumented incidences of fraud activities.

The combination of blockchain is another factor that makes the system different compared to the conventional methods. Auditable logs Compliance events in traditional systems are usually recorded in central databases, which are susceptible to manipulation or post-factum alteration. Using a smart contract on a test network of Ethereum on a permissioned system, these transactions are now permanently dropped on a blockchain with pertinent side data that includes risk score, origin of flag, and time stamp. This offers an auditable and transparent path that can be used for scrutiny by the regulatory branch.

In general, the unified AI and blockchain platform transforms the compliance monitoring process, which used to be periodic, into an ongoing process. It raises transparency, limits latency and has cryptographic-quality traceability, an immense advance over the current manual or semi-automated approaches.

5.5 Expert Feedback or Usability Testing

Given the secondary data-driven approach, no formal user testing has been conducted as regards to the financial professionals. Nevertheless, technical feasibility and usability of the system outputs and functions were analysed by peers and academic supervisors. The feedback was collected at the end of iterative sprint reviews, as it coincides with the ***Agile methodology***.

Users were glad to see the visibility of the results of the model, in particular, the classification reports and confusion matrices of each machine learning model were created. **Logistic Regression** was also seen to perform poorly in recall as compared to the **Random Forest**, which was able to achieve improved accuracy and **F1-scores** given class-imbalanced datasets. Isolation Forest was praised because it had limits to their supervision and that it helps to illuminate outliers, particularly in instances where there is limited labelling.

The cli_logger.py was assessed for ease of use. It performed the automatically flagged transactions, logged them in the blockchain, and came up with manual confirmation. The reviewers commented that it was a good modular design to have a separation of risk detection (black_money_model_evaluation.py) and blockchain recording (blockchain_logger.py) since it would be easy to extend the system and maintain.

The **Solidity** program (BlackMoneyLogger.sol) and tested on the Ganache project and indicated stable behaviour when it comes to logging the hash of transactions and related metadata. The logs on the output verified the irreversibility of the records, in addition to the ability to visualise the logs using the Ethereum interface block explorer.

Although the practical usability testing in a real-life financial environment is currently beyond the potential of this dissertation, the prototype currently developed does demonstrate the general operability and interaction logic that will be possible in the future to extend to the actual money-related application. Further development would benefit from the integration of graphical dashboards and a user role facility to promote the user experience and workflow of compliance officers further.

5.6 Discussion of Key Findings

The presented idea to create RegTech using artificial intelligence and integrating blockchain demonstrated three major lessons that could help in the future adoption of such technology in the context of compliance monitoring. On the one hand, with the help of machine learning models, especially **Random Forest**, the detection ability of suspicious transactions is high, as the accuracy and recall values are high. Logistic Regression served as a baseline to compare the results against, and Isolation Forest was used to provide robustness since it identified novel anomalies, whereas it did not depend on the availability of labels.

Second, risk identification through feature importance analysis built on ensemble models provided interpretability and traceability of the decision-making process regarding key transactional attributes that contributed to the decision. This strengthens the value of applying data-driven methodologies to increase regulatory clarity and weaken reliance on fixed rule-sets.

Chapter 6: Project Management

6.1 Project Planning and Gantt Chart

The project was to be implemented in a structured period of five months on a phased basis in line with the concepts of agile methodology. Each stage involved an iterative development cycle and included milestone-based monitoring of progress, which helped in the management of the progress. The planning process was done using a Gantt chart, to give an overview of the timeline and the dependencies and the distribution of the tasks, so that the AI model development and blockchain integration process go with the research objective of the project.

Month 1 was devoted to requirement analysis, literature review and acquaintance with the dataset. At the same time, the preprocessing of the Big Black Money Dataset was performed. It is during Month 2 that model development began, which included the use and testing of three classifiers, such as ***Logistic Regression, Random Forest, and Isolation Forest***, all in Python. The model performance was evaluated using ***ROC-AUC, precision and recall***.

The 3rd month was the implementation of this component of the blockchain. BlackMoneyLogger.sol, a Solidity-based smart contract, is written and deployed with the help of Ganache and connected using Web3.py scripts. The 4th month was devoted to system integration, i.e. connecting the anomaly detection software to the blockchain logger (blockchain_logger.py), and creating the CLI interface (cli_logger.py).

Last, System testing, expert opinion gathering, final assessment and report were done in Month 5. This was a formal method of doing the project in that it achieved technical and academic project output within the timeline proposed.

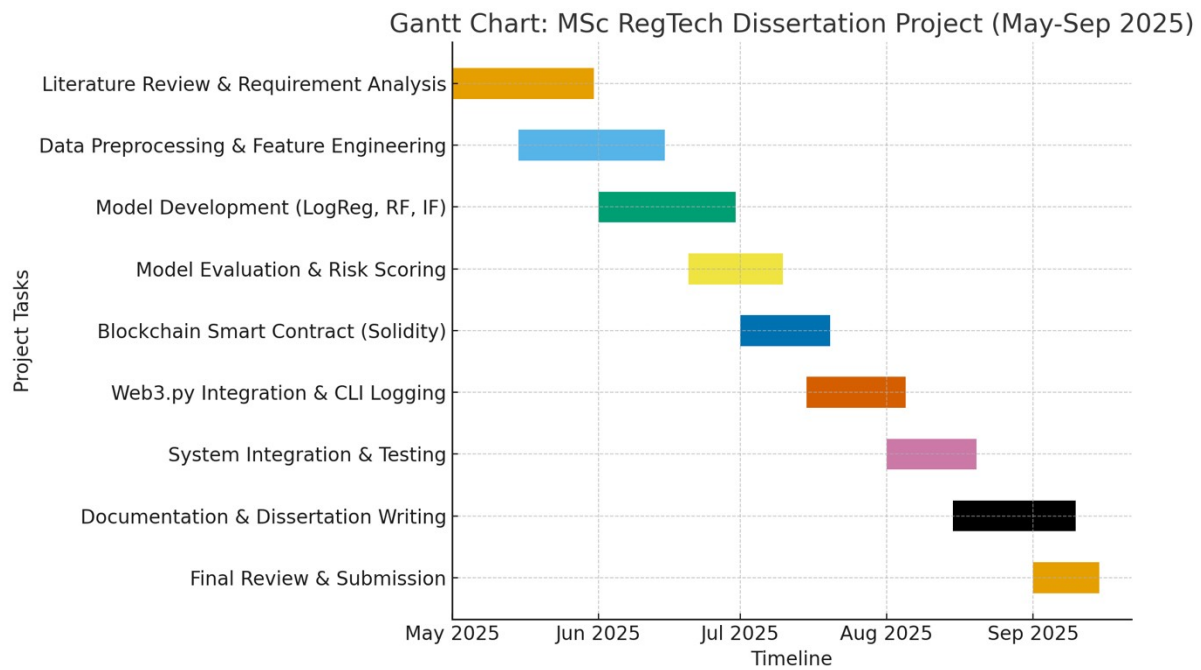


Figure 19: Gantt Chart of project evaluation

(Source: Self-created)

6.2 Resource Allocation

The resource deployment of this research project was phrased strategically in order to assist in the process of designing, integrating, and testing a **RegTech** prototype with **AI-based anomaly detection** and **blockchain-based audit-logging**. Since the research was conducted based on secondary data collection, it did not require any resources to collect primary data or recruit participants. Instead, the allocation had its roots in computational and software-based resources.

The main working environment was Python, in which we used libraries like scikit-learn to perform machine learning, pandas and NumPy to manipulate the data, and Matplotlib and seaborn to visualise the results. A high-performing local CPU, with at least 16GB of RAM and a multi-core processor, was utilised to efficiently run model training and evaluation tasks. In the development of blockchain, Ganache was adopted as an on-premise Ethereum test environment, and Solidity was adopted to develop the BlackMoneyLogger.sol smart contract. The interaction with blockchain was provided with the help of the web3.py library, which operates under the interfaces blockchain_logger.py and cli_logger.py.

Besides these dependencies, packages like **Remix IDE** and **MetaMask** to perform deployment of contracts and management of blockchain accounts, respectively, were also included as part of simulating a secure, permissioned network. This configuration guaranteed that identified transactions by the AI model could be permanently registered to the blockchain, therefore, providing the compliance audit trail mechanism at the core of the project.

6.3 SDLC and Agile Implementation

The system development has a hybrid approach where it has followed the model of SDLC and **Agile** to have the flexibility and systematic development. The **SDLC model** was used to give an organisational framework to go through the lifecycle stages of the **RegTech** prototype-requirements gathering, system design, implementation, testing and deployment and Agile was used to conduct iterative development, swift prototyping and continuous stakeholder feedback.

During the initial planning phase, the business and technical requirements were identified on the basis of regulatory compliance use cases and data characteristics derived from the **Big Black Money Dataset**. The modelling phase concerned both architecture modelling of the AI-based risk detection module and the Ethereum smart contract layer that would perform the blockchain logging. It was implemented in sections, in the form of Agile sprints, to provide deliverable functionality, such as the `black_money_model_evaluation.py` file, which solves the machine learning problem (Johnson., 2025). Additionally, `blockchain_logger.py` and `BlackMoneyLogger.sol` were developed to achieve reasonable performance and security on a private Ethereum network.

Analysis and testing were performed in an iterative manner, where models were trained and tested on secondary data, and blockchain functionality was verified using Ganache. In collaboration with the shell interface, the **CLI interface** (`cli_logger.py`) was iterated throughout the sprints to make it more user-friendly and to exclude unnecessary steps of interacting with a smart contract. This dual approach of traceability, flexibility and stability in scaling up and adapting a scalable, compliant, and technically robust **RegTech** solution.

6.4 Risk Management and Mitigation

The management of risks played a decisive role in the successful delivery of the proof of concept of the AI-Blockchain RegTech due to the high reliance on secondary datasets and

manual integration of custom codes. The main identified risks were inconsistency in data quality, the possibility of certain bias in the algorithms, failure to load a blockchain transaction, and difficulty in integrating the machine learning and the smart contract modules.

Therefore, to mitigate possible exposure to data-related risks, the project involved sound preprocessing measures including but not limited to the numerical scale matching, confidence category encryption, and outliers identification. This made the input in the entire machine learning models, **Logistic Regression**, **Random Forest**, and **Isolation Forest** standardised and representative. There were controls to reduce the risk of overfitting the models or biasing the model since the cross-validation was done in the strata, and the performance was evaluated using multiple metrics, including **ROC AUC** and **F1-score**, which improved generalizability across unseen data.

The blockchain aspect of it was that possible transaction failure could occur during the network turbidity or due to a mistake in contract deployment, but through experimentation on a local Ethereum test network (**Ganache**), all smart-contract-related interactions were tested (Chatterjee., 2025). The Python-based blockchain logger module is designed to include exception handling where the failure of transactions is noted and not terminating the system.

Lastly, configuration was modular, and the CLI tool was isolated to avoid system-wide failure and prevent lateral fixes. A combination of such mitigation strategies provided stability, transparency and robustness of the compliance automation framework.

6.5 Summary

This chapter gave insight into the project management approaches being undertaken in the development of the system, such as planning, implementation as well and mitigation of risks. AI-based risk detection and blockchain logging could be iteratively improved with the help of the agile methodology. A script written in Python was used to process secondary transactional data to train and test machine learning models, whereas **Solidity** and **Web3.py** were used in the realisation of the blockchain audit trail. The deployment was performed through the **CLI** and offered an expedited interaction with a contract and guaranteed transparent and tamper-resistant audit logs of high-risk transactions. High alignment of project activities to objectives made the delivery, functional integration and regulatory relevance of the RegTech prototype timely.

Chapter 7: Conclusion and Future Work

7.1 Summary of Research and Key Findings

This paper examined the design of a Proof-of-Concept **Regulatory Technology (RegTech)** which would automatically perform financial compliance checks using both machine learning as well as blockchain-based architecture. A secondary dataset of anonymised transactional records that may be related to money laundering and illicit financial activity served as the basis of analysis, with supervised and unsupervised learning being applied to recognise anomalous/high-risk transactions suggestive of non-compliance or money laundering.

The machine learning part of the system was done using Python, which has incorporated the following models: **Logistic Regression, Random Forest and Isolation Forest**. Analysis was performed on the preprocessed dataset by encoding categorical data and deleting noise. The effectiveness in using the models was measured using **accuracy, precision, recall and F1-score** performance metrics. The Random Forest model would be the best model to predict the target because it was able to capture non-linear trends of the transactional variables (Sosa and Sosa., 2025). The application of **Isolation Forest** in the unmonitored environment was useful in determining the outliers without the use of labelled data to complement the supervised models.

As a result of risk classification, those transactions that surpassed the risk threshold were passed to a permissioned blockchain smart contract written on **Solidity**. The smart contract, run on a local Ethereum test net via Ganache, offered immutable, auditable logging through the `logTransaction()` method, which deposited transaction identifiers, risk scores, source labels, and Unix timestamps in an externally-examinable on-chain structure. The connection of the AI detection module to the blockchain was realised by using the **Web3.py library**, which made it possible to execute logging events quickly and conveniently through Python.

The **CLI interface** also increased usability, which means that manual or automated submission of flagged transactions could be done quickly. Together, automated risk scoring and decentralised audit trails made real-time monitoring and reporting of compliance, data integrity and transparency to regulatory bodies. The system provides evidence on the viability of the combination of AI and blockchain in the RegTech setting and can be a prospective model that can be further studied and used in the operation of financial institutions.

7.2 Research Contributions

In this work, there has been a lot of contribution to the high-tech field of regulatory technology, whereby the practical natures of the application of machine learning and blockchain to automate financial compliance were simulated. By using secondary data (i.e., anonymised transaction histories), this paper designed and compared supervised machine learning (i.e., **Logistic Regression, Random Forest, and Isolation Forest**) algorithms in identifying possible illicit financial transactions. These models were applied to a Python-based analytical pipeline allowing efficient classification of anomalous transactions, providing real-time risk scoring with quantifiable performance measures including **precision, recall** and **ROC-AUC**.

The main technical contribution is the design and implementation of a smart contract, which was implemented in **Solidity** and which has the property of permanently storing flagged transactions on a permissioned Ethereum blockchain (Khang *et al.*, 2025). This component maintains auditable events, as transaction ID, risk score, and event time are recorded through the `logTransaction()` method via **Web3**. The integration of the blockchain increases the traceability and reliability of the compliance events that had limitations with traditional audit logs.

Additionally, the study proposes a replicable, **CLI-integrated** prototype that connects AI-powered risk analysis and blockchain-powered record-keeping. This end-to-end system architecture, powered by modular code and open-source tools, is scalable and audit-ready, and provides a basis for future development in automated financial compliance systems.

7.3 Limitations

Although the execution of the AI and blockchain-based RegTech prototype has proven to be viable in terms of functionality, we have to note several limitations. To start with, the secondary data has limitations for generalising. The use of the dataset given can be substituted with the real-world examples of the transactions of the bank system, but with various degrees of complexity and exceptions arising. Moreover, class imbalance typical of datasets used in fraud detection could have affected model sensitivity towards the minority (suspicious) classes and lowered predictive accuracy correspondingly, despite the use of resampling and algorithmic modifications.

Modelling-wise, the system combines **Logistic Regression**, **Random Forest**, and **Isolation Forest**, but this aspect fails to make dynamic thresholds to detect anomalies, which may not adapt well in the real world of fraud patterns. Also, the explainability of such models (ensemble-based) is relatively scarce, which may hamper regulatory transparency and auditability.

The blockchain portion of this application is built on a local **Ethereum test network (Ganache)** and a Solidity-based smart contract, which does not fully reflect the reality of a production-grade blockchain implementation (Jariwala., 2025). Gas cost, network latency, and privacy regulation limitations were modelled but not observed in a real sense. Lastly, the command-line-based integration is operational, although it does not have a friendly interface, which could disincentive non-technical compliance officers to incorporate the integration into the actual workflows in banks.

7.4 Future Recommendations

Although the RegTech prototype under development is an effective combination of supervised machine learning with blockchain technology in detecting real-time risks and compliance logging, some improvements are required in future investigations. To begin with, the data applied was secondary source data, and it might not completely reflect the complexity of live transactions in regulated banking environments. Future research ought to use more geographically and specialised datasets, ideally augmented with metadata related to geopolitical risks or multi-party transactions or layered financial instruments.

Machine learning, when assessing machine learning, there are advanced techniques that could be implemented into model integration that tend to increase soundness and precision, especially to pick up any partially hidden patterns of fraud. Also, the integration of **explainable AI (XAI)** components would facilitate more interpretable and regulatory auditability of decisions.

On the blockchain level, the implementation of a smart contract aims at static recording. This can be expanded in the future to utilise oracles to communicate with external compliance feeds or the use of zero-knowledge proofs to provide data privacy without interfering with its transparency (Jung., 2021). Integration with permissioned enterprise Blockchain like Hyperledger Fabric can also aid in good scaling and interoperability with pre-existing banking systems.

In addition, an automated connection between the anomaly detection pipeline and the blockchain logger can create a fully autonomous RegTech environment that eliminates latency in the flagging, recording of and responding to compliance violations in real-time.

7.5 Final Remarks

The present dissertation has succeeded in showing that the combination of supervised machine learning and blockchain technology can be used to automate financial compliance. Predictive models, created through secondary data analysis including Random Forest, Logistic Regression, and Isolation Forest, were used to identify high-risk transactions. These were efficiently integrated with a blockchain infrastructure based on smart contracts that guarantee the existence of an immutable audit trail and greater transparency. The Web3 interface and Python-based miner tool verified, in real-time, the logging of events of a permissioned Ethereum network. In general, the system demonstrates the ways in which the RegTech solution can perform with volumes beyond the approaches traditionally in use, with secure and explainable techniques applicable to the changing regulatory landscape of financial institutions.

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